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Benchmarking Brain Computer Interface for Motor Imagery

Tesi di laurea

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*To my grandmother,
Li, for always believing
I could achieve great
things...*

*Even if I started them
five minutes before the
deadline*

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Abstract

Brain–Computer Interfaces (BCIs) based on motor imagery (MI) hold promise for enabling direct communication between humans and machines, yet their practical deployment remains limited by low signal-to-noise ratio, non-stationary dynamics, and high inter-subject variability in EEG recordings. This thesis addresses these challenges by benchmarking classical machine learning models against modern artificial neural networks (ANNs) using the BCI Competition IV-2a dataset. Two complementary experiments were conducted: the first established a baseline with foundational methods such as filter-bank common spatial patterns (FBCSP) combined with feature selection and linear discriminants; the second evaluated advanced deep learning approaches, including convolutional and recurrent neural networks, to assess their ability to capture spatial and temporal EEG dynamics. Results reveal that while ANNs achieved superior performance in terms of balanced accuracy and Cohen’s κ , foundational pipelines remain competitive due to their efficiency, interpretability, and robustness with limited data. The analysis highlights the trade-off between complexity and generalization, underscoring both the opportunities and limitations of current methods for real-world BCI deployment. Beyond the empirical findings, this work provides a systematic benchmark and contributes to the understanding of how statistical learning and deep learning techniques can be integrated to advance MI-based BCIs.

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Chapter 1

Introduction

The development of functional Brain-Computer Interface (BCI) technology represents a crucial and highly engaging frontier in biomedical engineering and neuroscience. The promise of restoring motor function through mental volition, particularly via the Motor Imagery (MI) paradigm, is remarkable, yet its widespread clinical deployment is consistently challenged by the fundamental hurdle of achieving robust, real-time signal classification.

This thesis directly addresses this critical challenge. With the rapid evolution of deep learning and the proliferation of powerful, generative AI models, a systematic re-evaluation of classification pipelines for MI data is not just beneficial—it but necessary.

In this endeavour, the work presented here makes a step further by undertaking a rigorous, comparative study.

By establishing an optimal framework that systematically explores preprocessing techniques and benchmarks the performance of both classic and cutting-edge Artificial Intelligence (AI) / Machine Learning (ML) models, this research will provide actionable insights into developing the next generation of highly accurate and reliable BCI systems. The findings presented here aim at closing the gap between laboratory research and clinical reality in the pursuit of effective neuroprosthetic control, and BCI systems at large.

1.1 Research Objectives

1.1.1 Motivation and Context

BCI applications face several problems that are in need of solving. Inside the specific MI realm, there is the online or real-time classification of a continuous signal [Tangermann et al. \(2012\)](#).

This is not a new problem by any measure, but with the advent of Generative AI and models designed to compete with the state-of-the-art GPT models, a benchmarking of classification models was due.

1.1.2 Research Questions

1. **For Motor Imagery, is there an optimal preprocessing technique for both ML and AI?**

MI Classification faces three main challenges: 1. Low Signal-to-Noise-Ratio (SNR), 2. Non-stationary nature of recorded signals, and 3. Variability within subjects. The different techniques for preprocessing, grounded in a long tradition of signal processing should be explored, in accordance with the literature, in order to determine the most efficient and accurate ones.

2. **Which foundational Machine learning models are the best suited ones for classification tasks in Motor Imagery?**

For this question an research and experimental approach will be followed to determine the best models in this context.

3. **Are CNNs suited for the BCI-MI Classification problem, and what constraints do they face?** To answer this, the experiment and pipeline design must be well researched and tested. Conclusions reached after two different experiments carried out.

4. **Are RNNs suited for the BCI-MI Classification problem, and what constraints do they face?** To answer this, the experiment and pipeline design must be well researched and tested. Conclusions reached after two different experiments carried out.

1.2 Overview of Thesis

This Thesis unfolds as follows:

- **Chapter 1 - Introduction:** This chapter explains the context of the problem undertaken in the thesis, motivations for the experiments and research questions to answer.
- **Chapter 2 - Theoretical Background:** This chapter reviews the existing literature on Motor Imagery (MI), EEG signal processing, feature extraction, and machine learning models for MI classification, as well as Artificial Neural Networks.
- **Chapter 3 - Materials and Methods:** This chapter describes the dataset used in the study (BCI IV - 2a), the preprocessing techniques applied, feature selection, and the machine learning and Neural Network models employed.
- **Chapter 4 - Experimental Results:** This chapter explains the design of the two experiments performed, and then discusses the result.
- **Chapter 5 - Discussion and Conclusions:** The findings are analyzed in depth, also this final chapter summarizes the key findings, highlights the contributions of the study, and discusses the limitations. Future research directions, including potential extension to other datasets, model refinement, and more testing.

Chapter 2

Theoretical Background

This chapter provides a comprehensive overview of the theoretical and methodological foundations underlying EEG-based Motor Imagery Detection. The overview begins with the Electroencephalography (EEG), and its role as the sole source of data for this study. The fundamentals of EEG acquisition and preprocessing and the types of features commonly extracted from EEG signals. Then with that background established, the framework will be expanded to Motor Imagery (MI), and ways in which the EEG data is applied to MI problems via machine learning models used for classification, and finally, the application of Artificial Neural Networks to this classification problem.

2.1 Electroencephalography (EEG)

Electroencephalography (EEG) is a neuro-imaging technique that enables real-time recording of the brain's electrical activity. It works by measuring changes in electrical potential in the scalp, caused by synchronous activation of neighboring neurons (Ruff & Huettel (2014)).

The EEG signal represents the summation of electrical activity generated by several thousands or millions of neurons, typically recruited to perform a cognitive task. It originates primarily from the postsynaptic potentials of pyramidal neurons, which are located in the cerebral cortex and oriented perpendicularly to the cortical surface. When a sufficient number of these neurons fire synchronously, the resulting electrical fields aggregate and can be detected on the scalp using electrodes.

EEG signals are characterized by very low amplitudes, typically ranging between 10 and 100 μV , and must be amplified prior to analysis. The acquisition process involves several key steps:

1. **Signal detection:** Electrodes placed on the scalp capture voltage differences between different cortical regions.
2. **Amplification:** Since cerebral electrical activity is weak, the signal is amplified to enhance its quality.

3. **Digitization:** The amplified signal is converted to digital form via an analog-to-digital converter (ADC). The sampling frequency typically ranges between 250 Hz and 1000 Hz, depending on the required resolution.

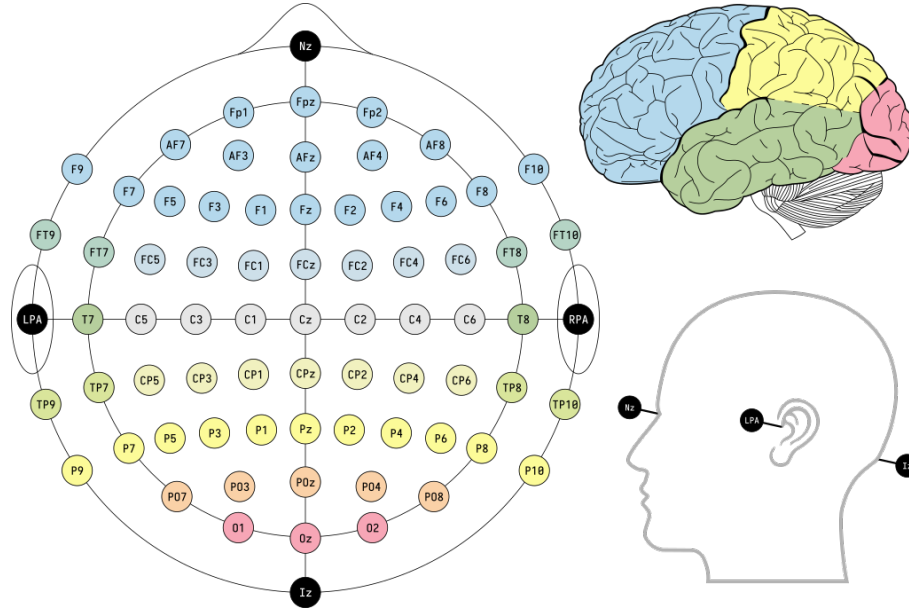


Figure 2.1: 10-20 International System

To ensure standardized and comparable recordings across different subjects, electrode placement follows international conventions. The most widely used system is the **10-20 International System**, which derives its name from the fact that electrodes are positioned at distances corresponding to 10% or 20% of the total length of specific anatomical reference lines on the skull. This system ensures a uniform electrode distribution across the cortical surface, allowing for reliable measurements. Electrodes are positioned in specific locations identified by letters corresponding to the cortical area they monitor:

- **F** (Frontal) → Involved in cognitive and executive functions.
- **C** (Central) → Associated with motor control and sensory processing.
- **P** (Parietal) → Responsible for multi-sensory integration and spatial perception.
- **O** (Occipital) → Primary region for visual processing.
- **T** (Temporal) → Involved in auditory processing and memory.

Graphical Sketch is shown in Figure 2.2.

Each electrode is assigned a number: even numbers indicate placement in the right hemisphere, while odd numbers correspond to the left hemisphere. Electrodes along the midline are designated with the letter *z* (e.g., Fz, Cz, Pz, Oz), ensuring a precise alignment with the subject's cranial morphology.

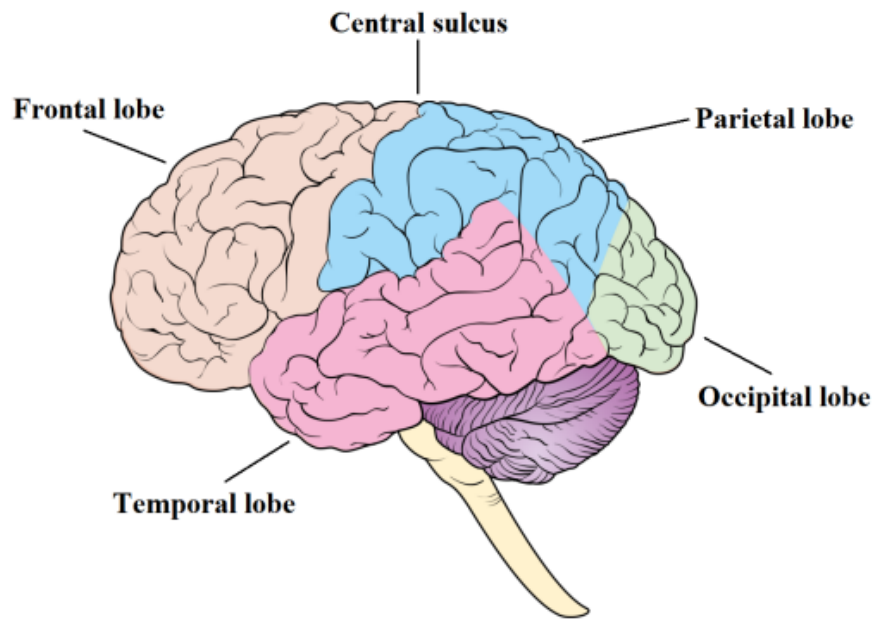


Figure 2.2: Anatomy of Human Brain

EEG is particularly valuable for studying dynamic variations in brain activity due to its **high temporal resolution**, making it an essential tool for investigating cognitive processes, states of consciousness, neurological disorders, and neural responses to external stimuli.

Another significant advantage of EEG is its **non-invasiveness**. Unlike other neuro-imaging techniques such as functional magnetic resonance imaging (fMRI), which requires controlled environments and may be uncomfortable for some subjects, EEG allows measurements without exposure to radiation or contrast agents. This characteristic, combined with its relative ease of use and portability of EEG devices, makes it an accessible experimental method that can be used in various settings, including experimental studies conducted outside of traditional laboratory environments.

However, EEG also has certain limitations that must be considered when interpreting data. One of the primary drawbacks is its **low spatial resolution**, as the signal must traverse multiple layers of tissue (such as the skull and scalp) before reaching the electrodes, making it difficult to pinpoint the exact origin of the recorded brain activity. This limits EEG's ability to precisely identify cortical regions involved in specific cognitive functions or responses to stimuli.

Another challenge is its **sensitivity to artifacts**, which in other words are unwanted signals that can contaminate recordings. Eye movements, muscle activity, and electrical interference from the environment are all factors that can distort EEG data, necessitating a rigorous *preprocessing* phase to enhance signal quality. Despite these limitations, EEG remains an effective technique for studying neural

responses to external stimuli and monitoring variations in neuronal activity related to cognitive and emotional states. Its ability to provide a direct measure of brain electrical activity and track real-time neural dynamics makes it particularly suitable for investigating stress perception and its neurophysiological consequences.

2.1.1 Time-Domain Analysis of EEG Signals

For the specific analysis and processing of EEG, it is taken as a discrete-time stochastic process $x[n]$ obtained by sampling a continuous bioelectrical signal. It is non-stationary (meaning that its statistical properties change across time) at long scales but often treated as **quasi-stationary** within short epochs, enabling time-domain processing on segments $x_w[n] = x[n] * w[n]$ defined by a window $w[n]$.

Linear Time-Invariant filtering in time. Time-domain filtering is expressed as a convolution (denoted with $*$) with an impulse response $h[n]$:

$$y[n] = (h * x)[n] = \sum_{k=-\infty}^{\infty} h[k] * x[n - k]. \quad (2.1)$$

Finite-impulse-response (FIR) and infinite-impulse-response (IIR) filters are written as

$$\text{FIR: } y[n] = \sum_{k=0}^L b_k x[n - k], \quad (2.2)$$

$$\text{IIR: } y[n] = \sum_{k=0}^M b_k x[n - k] - \sum_{m=1}^A a_m y[n - m], \quad (2.3)$$

and are typically applied in a *zero-phase* manner (forward-backward) to avoid phase distortion in event-locked analyses.

Time-domain statistics and features. On an epoch of length N , common features are computed directly on the samples:

$$\mu = \frac{1}{N} \sum_{n=1}^N x[n], \quad \sigma^2 = \frac{1}{N} \sum_{n=1}^N (x[n] - \mu)^2 \quad (2.4)$$

For multichannel data (channels $\times$ time), the element-wise *sample covariance* is:

$$S_{ij} = \frac{1}{T-1} \sum_{t=1}^T (x_{i,t} - \mu_i)(x_{j,t} - \mu_j) \quad (2.5)$$

These matrices quantify spatial amplitude co-fluctuations in the time domain.

Common Spatial Patterns (CSP)

CSP in the time domain. Common Spatial Patterns (CSP) operates on time-domain covariances. For two classes with class-wise averages Σ_1, Σ_2 (often shrunk with Oracle Approximation Shrinkage (OAS)), CSP solves the generalized eigenproblem: separating a multivariate signal into additive subcomponents which have **maximum differences in variance** between two windows.

$$\Sigma_1, w = \lambda, (\Sigma_1 + \Sigma_2), w, \quad (2.6)$$

$$f_k = \log! \left(\text{var}(Z_k) \right), \quad k = 1, \dots, m. \quad (2.7)$$

In filter-bank CSP, this procedure is repeated on band-pass-filtered *time* signals, and features are concatenated.

Oracle Approximating Shrinkage (OAS): is a computationally fast, closed-form method to regularize (in this context smooth and remove noise to make them somewhat well-behaved) sample covariance matrices. Typically used for when there are many channels in an experiment. It focuses on shrinking the sample covariance Mean Square Error (MSE) vis-à-vis the true population covariance. It uses a scaled identity: α to stabilize inversions/eigen-decompositions:

$$\hat{S} = (1 - \alpha)S + \alpha sI \quad (2.8)$$

where s is the scalar mean variance per channel used to scale the identity. That is, a scalar value equal to the average eigenvalue or variance per channel, calculated as the trace of a matrix divided by the number of channels:

$$s = \frac{1}{p} \text{tr}(S) \quad (2.9)$$

OAS shrinks toward the spherical target, preserving the overall power (same trace) while improving conditioning.

2.1.2 Frequency-Domain Analysis of EEG Signals

For spectral analysis, the discrete-time EEG process $x[n]$ is transformed into the frequency domain, providing a representation of how signal power is distributed across oscillatory components Oppenheim et al. (1996). Because EEG activity is composed of multiple overlapping rhythms (δ , θ , α/μ , β , γ), this view reveals the relative contribution of each frequency band to brain dynamics.

In practice, $x[n]$ is segmented into quasi-stationary epochs (i.e. epochs with a "short-enough" time window) and each segment is transformed using a discrete Fourier transform (DFT):

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n} \quad (2.10)$$

The resulting complex spectrum $X[k]$ encodes both amplitude and phase for frequency bin $f_k = kf_s/N$, where f_s is the sampling rate. The corresponding **power spectral density (PSD)** is defined as:

$$S_{xx}(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |\hat{x}_T(f)|^2 \quad (2.11)$$

which estimates the signal’s energy distribution over frequency. Linear filtering in the frequency domain exploits the convolution theorem, where time-domain convolution corresponds to frequency-domain multiplication.

Band-pass or notch filtering can therefore be implemented equivalently in the spectral domain by zeroing or attenuating specific frequency bins. Frequency-domain features such as band power ratios, spectral entropy, or peak frequency are widely used in motor-imagery BCI, as event-related desynchronization (ERD) and synchronization (ERS) phenomena manifest as task-dependent power changes in the μ (8–12 Hz) and β (18–26 Hz) bands.

2.1.3 Time–Frequency Analysis of EEG Signals

EEG signals are inherently nonstationary: their spectral content changes rapidly across time, particularly in response to sensory events or during motor imagery tasks. To capture these transient spectral dynamics, **time–frequency analysis** combines temporal and spectral perspectives, representing the signal energy as a joint function of time and frequency. This is where, the Short-Time Fourier Transform (STFT), an instance of **Windowed Fourier Transform**, is useful.

Short-Time Fourier Transform (STFT)

The **short-time Fourier transform (STFT)** provides a localized frequency representation by applying the Fourier transform to successive, overlapping windows (chunks) of the signal. Each windowed segment $x_w[n] = x[n]w[n - m]$ is analyzed separately, producing the complex time–frequency representation

$$STFT\{x[n]\}(m, \omega) \equiv X(m, \omega) = \sum_{n=0}^{N-1} x[n]w[n - m] e^{-i\omega n} \quad (2.12)$$

The squared magnitude yields the **spectrogram**, $Spectrogram\{x(t)\}(\tau, \omega) \equiv |X(\tau, \omega)|^2$ which reflects how power in different frequency bands evolves across time. The choice of window $w[n]$ determines the trade-off between temporal and frequency resolution: shorter windows capture rapid temporal changes with coarse spectral detail, while longer windows provide sharper frequency estimates at the cost of temporal precision.

As stated by C. & Chui (1993), a drawback of both Fourier series and Fourier transforms is that they destroy local information. This is particularly relevant in the context of EEG, and time-sensitive cued MI. In an effort to also get a hold of local information, the windowed Fourier transform takes the Fourier transform of many localized versions of the original signal. The resulting STFT does give local information about the signal $x[n]$, at the expense of introducing a ‘position’ parameter m . As shown in, 2.3, the result is then a representation of the amount of frequency present in the graph of f intersection the window w around m .

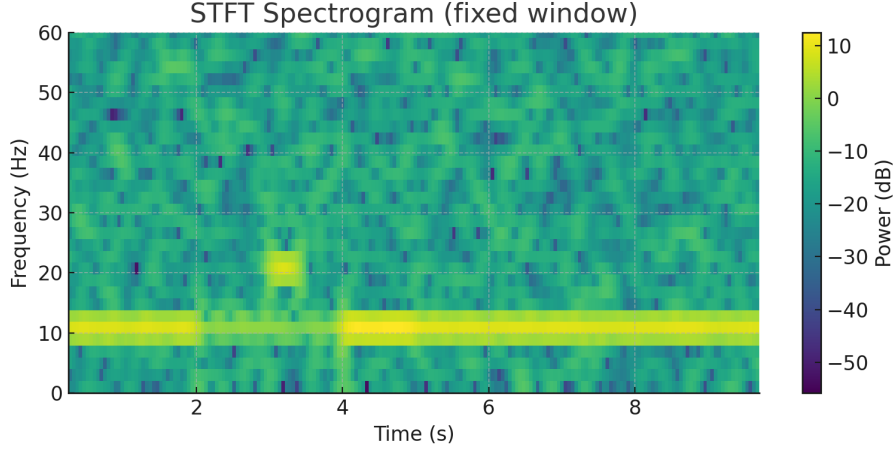


Figure 2.3: Anatomy of Human Brain

Wavelet Transform (WT)

For a relatively high frequency ω where the corresponding period $1/\omega$ is smaller than the window, we are in a reasonably good position to detect whether there is a frequency ω component inside the window. For relatively small frequency, and large period, we do not really have a big enough window on the graph of f to say that we are detecting any variation in the graph at that frequency. This suggests that if we want to know where we can find a component of frequency ω in the graph of f , we should not expect to be able to answer unless we can consider a section (or window) of the graph of length comparable to the period $1/\omega$.

$$W(f)(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} \overline{\psi\left(\frac{t-b}{a}\right)} f(t) dt \quad (a > 0, b \in \mathbb{R}) \quad (2.13)$$

where a is the *scale* (inversely related to frequency) and b is the time translation. Small scales correspond to high frequencies (fast oscillations), while large scales capture slower rhythms. Mapping scales to their corresponding pseudo-frequencies $f_a = \frac{f_c}{a}$ (with f_c the central frequency of ψ) allows the wavelet coefficients $W(a, b)$ to be visualized in time–frequency form, analogous to a spectrogram but with adaptive resolution—fine temporal detail at high frequencies and fine frequency detail at low frequencies.

Applications to EEG and Motor Imagery

Wavelet transform is an alternative, multi-resolution approach, which decomposes the signal into dilated and shifted versions of a mother wavelet $\psi(t)$.

In EEG analysis, time–frequency methods are particularly powerful for studying **event-related desynchronization (ERD)** and **event-related synchronization (ERS)**—the transient suppression and rebound of rhythmic activity within specific bands (typically μ : 8–12 Hz, and β : 18–26 Hz) following a motor cue or during

imagined movement.

2.1.4 Event-related Analysis

EEG by itself is powerful in assessing the overall state of the brain. However, it is limited in its ability to study specific cognitive functions because it is not linked in time to any particular phenomena as per Pfurtscheller et al. (2006). Thus, a pairing of time of stimuli and the study of the event-related changes in the brain processes has been introduced.

Event-related analysis. For time-locked responses (ERPs), per-trial epochs $x_i[n]$ are baseline-corrected and averaged:

$$\bar{x}[n] = \frac{1}{M} \sum_{i=1}^M x_i[n], \quad (2.14)$$

emphasizing stimulus-locked deflections while suppressing uncorrelated noise—again entirely in the time domain. Although no explicit Fourier transform is computed here (hence, no Frequency-domain analysis), band-pass filtering plus log-variance features effectively capture *band power* thanks to Parseval’s theorem; the analysis remains time-domain in implementation and interpretation.

Event-Related Potentials: time-locked averaging approach connected to outside stimuli directly related to the cognitive process or event we want to study Purves et al. (2013). ERPs are small voltage fluctuations directly triggered by sensory and cognitive events (cues).

Event-Related Synchronization (ERS): relative rise in oscillatory power in a band due to those neurons being recruited for a cognitive task (synchrony).

Event-Related Desynchronization (ERD): relative drop in oscillatory power in a band. Reflects reduced synchrony of local neuronal populations.

2.1.5 Brain Waves

The EEG signal is composed of multiple oscillatory components, categorized into frequency bands based on their speed and functional significance. These are also known as Sensorimotor Rhythms, each band is associated with specific cognitive processes and states of consciousness.

Theta Waves (4 – 8 Hz): Theta oscillations are characterized by low frequency and are typically associated with deep relaxation states, meditation, and memory-related cognitive processes. These waves are particularly prominent in the **temporal and frontal regions**, with strong expression in the **hippocampus**, a key structure for memory and learning.

From a functional perspective, theta waves are primarily observed in:

- Transition states between wakefulness and sleep (Stage 1 and 2 of sleep), where they facilitate the processing of information acquired during wakefulness.

- Learning and memory processes, especially within the hippocampal regions, where they contribute to memory consolidation.
- Deep meditation and introspection states, showing increased activity during practices involving heightened self-awareness.

Increased theta power has been associated with enhanced **creativity** and problem-solving processes, whereas a reduction in theta activity has been linked to learning disorders and working memory deficits.

Alpha Waves (8 – 13 Hz): Alpha waves are typically associated with states of relaxed wakefulness and are considered a marker of cortical inhibitory activity. These oscillations are most prominent in the **occipital and parietal regions**, where they dominate when an individual is awake but has their eyes closed.

From a functional perspective, alpha waves are involved in:

- Mental relaxation states, occurring in the absence of demanding cognitive tasks, indicating a balance between neural excitation and inhibition.
- Cortical inhibition mechanisms, regulating the transition between wakefulness and attentional states, facilitating focus on relevant stimuli while suppressing irrelevant information.
- Decreased power in states of stress and anxiety, where the prefrontal cortex exhibits increased beta activity.

Mu Waves (8 – 13 Hz): Both α and μ waves are defined over the same frequency. α is strongest over occipital/parietal cortex ($O_1/O_z/O_2$), while μ is strongest over sensorimotor cortex ($C_3/C_z/C_4$). Topography being the first main distinctive feature Marshall & Meltzoff (2011).

Reactivity: Processes which stimulate/dampen the waves

- Occipital α (“visual alpha”) grows with eyes closed and drops with eyes open/visual attention; it’s often interpreted as visual cortical inhibition (idling) Hohaia et al. (2022)
- Sensorimotor μ drops (ERD) with actual or imagined movement and with somatosensory input; after movement you can see a β rebound (>13 Hz) rather than μ -ERS. This pattern is central/contralateral and is the backbone of MI-BCI Marshall & Meltzoff (2011)

μ waves exhibit clear ERD (decrease in amplitude) over sensorimotor cortex during MI tasks.

Beta Waves (13 – 30 Hz): Beta waves are typically associated with states of alertness, problem-solving, and intense cognitive activity. They are predominantly observed in the **frontal and central regions** and play a role in regulating voluntary behavior and **motor activity**.

From a functional perspective, beta waves are involved in:

- Complex cognitive activities, such as problem-solving and action planning.
- **Motor regulation**, through interactions with motor cortex circuits and the basal ganglia.
- Increased presence in states of stress and anxiety, due to heightened vigilance and emotional response regulation.

β activity is usually observed in ERD alongside μ during MI. After actual motion (not MI), there is usually an observed rebound Khademi et al. (2023).

Gamma Waves (30 – 100 Hz): Gamma oscillations are the fastest brain waves and are involved in **neural synchronization** across different brain regions. These waves are associated with high-level cognitive functions, such as sensory processing integration, abstract thinking, and consciousness.

From a functional perspective, gamma waves are involved in:

- Complex cognitive processing, including reasoning, language, and abstract thought.
- Working memory and sensory integration across multiple cortical regions.
- High states of consciousness, including advanced meditation states and *flow* experiences.

Altered gamma wave activity has been observed in various pathological conditions. Reduced gamma power has been linked to **cognitive disorders**, such as **Alzheimer’s disease** and schizophrenia, highlighting their critical role in inter-regional brain communication. Some studies have also reported an increase in frontal gamma power under acute stress or intense cognitive demand. ? observed that relative frontal gamma increased significantly during the Montreal Imaging Stress Task, suggesting its potential as a marker for high cognitive load and stress.

2.1.6 EEG Preprocessing

As previously mentioned, EEG is highly susceptible to noise, and its raw form is often contaminated by **artifacts**, which are unwanted signals that do not originate from neuronal activity but rather from physiological or environmental interferences, such as eye blinking, muscle contractions and so on. This is due to the fact that EEG records electrical potential differences with very low amplitude, typically in the order of **microvolts** (μV), making it particularly vulnerable to contamination from both the body and the surrounding environment. If these artifacts are not removed, they can compromise signal quality and lead to incorrect data interpretation, making preprocessing a crucial step in ensuring accurate and reliable EEG analysis Purves et al. (2013).

Main EEG Artifacts

EEG signals can be affected by various types of artifacts, which can be classified based on their origin. The main categories of artifacts include:

Ocular Artifacts Ocular artifacts are among the most common and are caused by eye movements or the activity of the orbicular muscles. These artifacts typically appear as high-amplitude waves in the frontal regions and can be categorized as follows:

Slow eye movements and **blink artifacts**, which produce high-amplitude deflections that interfere with the analysis of brain waves, and **saccades**, rapid eye movements that introduce transient variations in the EEG signal. Since the eyes act as electrical dipoles, potential variations due to their movement propagate easily across the scalp, contaminating the EEG signal.

Muscle Artifacts Muscle artifacts (*electromyographic artifacts, EMG*) arise from the electrical activity generated by contractions of the facial, neck, and head muscles. These artifacts are characterized by relatively high frequency (> 30 Hz), overlapping with **beta** and **gamma** EEG waves. They are particularly problematic in high-frequency EEG studies, as their spectral distribution overlaps with that of gamma waves, making it difficult to distinguish true neural activity from EMG contamination.

Electrical Artifacts Electrical artifacts stem from environmental interferences and the quality of electrode contact with the scalp. The primary source of these artifacts is **powerline interference (50/60 Hz)**, which can contaminate EEG signals, especially in environments with nearby electronic devices.

Motion Artifacts Motion artifacts occur when the subject makes involuntary head or body movements, leading to variations in electrode positioning. These artifacts can cause high-amplitude fluctuations in the EEG signal and are particularly problematic in mobile EEG recordings.

Artifact removal requires the application of various filtering techniques, signal decomposition methods, and advanced statistical approaches, often varying depending on the type of artifact. The preprocessing methods presented here are specifically tailored to the dataset used in this study.

Band-Pass Filtering

Band-pass filtering is an essential step in EEG preprocessing, allowing the selection of a frequency range of interest while eliminating unwanted components. In the context of EEG, this type of filter is used to remove unwanted frequencies from the signal.

Mathematically, the ideal transfer function of a band-pass filter is defined as:

$$H(f) = \begin{cases} 1, & f_L \leq f \leq f_H \\ 0, & \text{otherwise} \end{cases} \quad (2.15)$$

where f_L and f_H represent the lower and upper cutoff frequencies, respectively. However, in practice, the filter response is not perfectly binary and features a gradual transition between the passband and attenuated frequencies.

For EEG signals, the commonly used filtering method is **Finite Impulse Response**

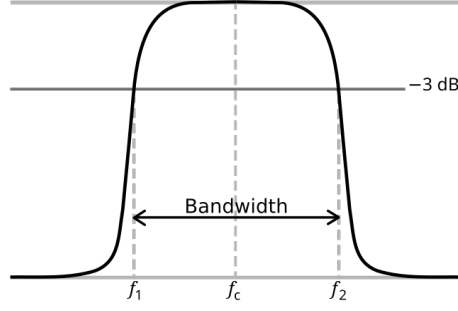


Figure 2.4: Transfer Function of a band-pass filter

(**FIR**), which provides a stable and predictable frequency response.

Using well-designed FIR filters ensures numerical stability and phase distortion-free filtering, enabling precise pass-band definition without altering the informative content of the EEG signal. This is crucial for extracting meaningful data and ensuring the reliability of subsequent EEG analyses.

2.2 EEG Features

Raw EEG signals contain a vast amount of information that must be processed through **feature extraction** to be effectively utilized in computational analyses. The goal of this process is to identify representative patterns of brain activity by extracting features that capture variations in neural activity associated with specific cognitive processes: Motor Imagery (MI) in the context of this thesis.

In EEG analysis, **signal segmentation** is crucial for extracting meaningful information from adequately sized temporal windows. The duration of the segmentation window is a critical parameter, as it determines the type of information captured within each segment. Different study suggest that for stress detection, **2-5 second** windows provide a good balance between temporal resolution and stability of extracted features.

EEG feature extraction plays a crucial role in translating raw brain signals into structured, discriminative representations for MI detection.

Khademi et al. (2023) provide an extensive review of MI-BCI, including the challenges mentioned in the Introduction Chapter of this thesis: 1. Low Signal to Noise Ratio (SNR), which is generally addressed with spatial filtering, 2. Non-stationarity and 3. Subject variability. These challenges can be undertaken with different algorithms.

In the context of low SNR, the most common solution is Spatial Filtering, among the most common methods, the aforementioned CSP is the method with the highest potential for generalization and widespread use as reported by Khademi et al. (2023) and Tangermann et al. (2012). The most commonly used feature categories in mental stress detection include time-domain, frequency-domain, time-frequency, nonlinear, and connectivity-based features. These feature types offer complementary information: while frequency-based features characterize the spectral distribution of

neural oscillations, nonlinear features capture the irregularity, complexity, and unpredictability of brain signals under stress.

Building on these insights, the present study adopts a time-domain approach, mimicking the top-performing state-of-the-art models from BCI IV competition and papers in the adoption of ANNs in MI-BCI. This approach aims to determine the most efficient and effective ways to implement ML and AI in the context of pipelining BCI MI.

2.2.1 Time-Domain Features

The two main approaches for in Feature engineering for EEG are Time-Domain and Frequency-domain analysis. Since EEG is a multi-channel signal, feature extraction is performed for each band, for robustness and to simplify analysis and computing efficiency.

Common Spatial Patterns (CSP)

This is a Spatial Filtering method, focusing on the band-limited time series. By creating an aggregate of channels, this new "space" helps filter signal strength and response to stimuli.

This approach implicitly handles, encodes and processes non-linear relationships in the data.

2.2.2 Univariate Filtering

An extra filtering step can be implemented, using a univariate scoring function. One common metric is Mutual Information between each feature and the class label Y , once the score is determined, one can rank features by that score, and retain the best k features. This method does not model feature interactions or redundancy; it is computationally fast and is typically placed inside an "inner" Cross-Validation step to avoid information leakage between train-test splits.

Mutual Information

Entropy is a measure of the degree of disorder and unpredictability of a signal. It allows to quantify the intrinsic complexity of the temporal dynamics of the signal itself.

For a discrete variable X , with pmf $p(x)$, entropy $H(x)$ is:

$$H = - \sum_x p(x) \log p(x) \quad (2.16)$$

Conditional Entropy: $H(X|Y)$ measures the remaining uncertainty in X after Y has been observed. **Mutual Information:** I is generally robust, and able to

account for non-linear and non-monotonic dependencies that a correlation test would miss. It is also model-agnostic. Its formula is:

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \quad (2.17)$$

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = D_{KL}(p(x, y) || p(x)p(y)) \quad (2.18)$$

Where if $I(X; Y) \geq 0$ iff X and Y are independent. So, Mutual Information $I(X; Y)$ in the context of the experiment used to create a ranking, and later only the K-best features (optimized through CV) are selected.

2.3 Motor Imagery (MI)

Lotze & Halsband (2006) states that motor imagery (MI) is the outcome of conscious intention of a movement, which is usually performed unconsciously during movement preparation. Meaning, it's the measuring of the conscious imagining of movement. Understandably, there is high overlap between motor execution and imagery. There are some key concepts that need either revisiting from the general brain physiology discussed above, or definition when new:

- **Motor Systems in the brain:** Purves et al. (2013) The cognitive processing for Motor Execution (and Imagery) is largely confined to the pre-motor, supplementary-motor and motor cortex. These correspond to the specific electrodes associated with sensorimotor processes ($C_3/C_z/C_4$). Lotze & Halsband (2006) points that MI recruits much of the motor network, albeit with a weaker β ERS.
- **Kinesthetic vs Visual Imagery:** Are two ways of conducting the imagery. Kinesthetic entails "feeling" the movement, while visual "pictures" is. Research shows more robust results in Kinesthetic trials Lotze & Halsband (2006).
- **Paradigms & task design:** Studies can be carried in a synchronous (cue-based) and asynchronous (self-paced) manner; most datasets focus on cues, as they can also be used to link the EEG data to the timed cues.
- **Class sets:** depending on the scope of the study, number of channels and other determining factors, the experiment can include any number of classes. Generally, due to the distorted representation of body parts in the motor cortex, as well as the distance between the regions that control them, the most commonly used sets are :L/R hand, feet, tongue, multi-class, depending on the study, experiments may include control or no-control states, for ease of prediction/simplicity of study.
- **Feedback modalities (visual, haptic/auditory):** Different studies implement different ways in which to cue their participants. These vary in a) The length of time between stimulus and reaction and b) brain regions engaged in the cognitive task.

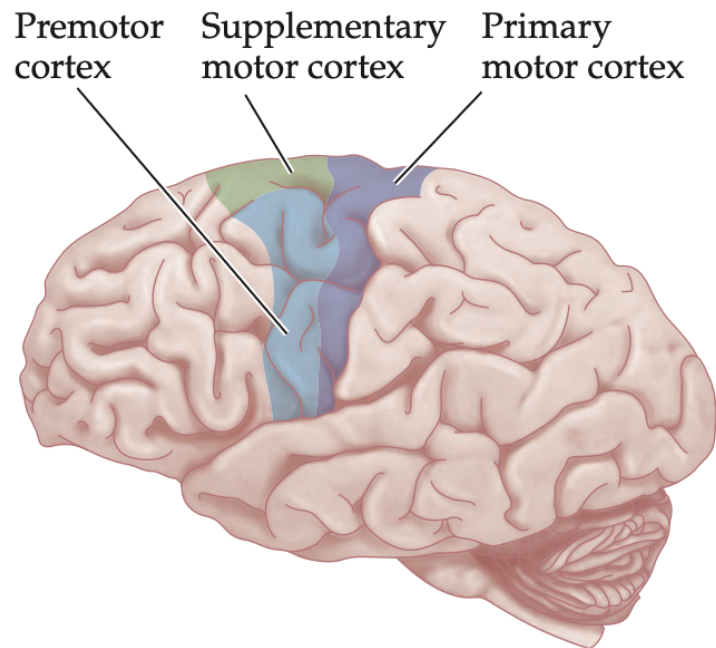


Figure 2.5: Primal Motor Cortex and Premotor Cortical areas

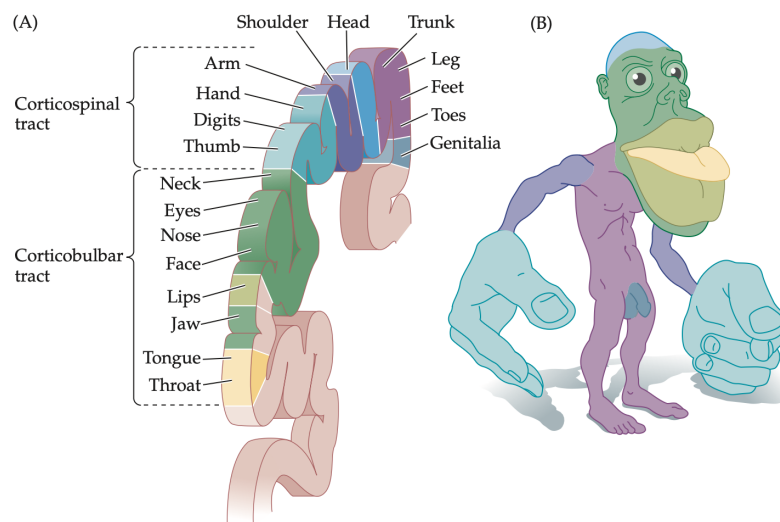


Figure 2.6: A. Regions in the Motor Cortex controlling different parts of the body and B. Distorted muscle representations in the Motor Cortex

- Event markers ; Event markers usually are used to link the spatial EEG data to the known (labeled) stimuli.
- Baseline selection: a pre-experiment fixation of the brain activity, in the same environment as the study, to estimate resting power.

2.4 Machine Learning Models

2.4.1 Case Study Complexity and Model Selection

The application of machine learning to EEG-based Motor Imagery classification is challenged by the limited size and heterogeneity of available datasets. As highlighted in the review by Tangermann et al. (2012), classical models such as Linear Discriminant Analysis (LDAs) and Support Vector Machines (SVMs) remain widely adopted due to their robustness and ability to generalize well with high-dimensional feature sets, even in low-sample contexts.

Based on these considerations, the present study adopts two widely used models for EEG-based classification:

- **Linear Discriminant Analysis.**
- **Support Vector Machine (SVM).**
- **Gaussian Naive Bayes Classifier (GNB)**

2.4.2 Linear Discriminant Analysis (LDA)

As discussed on Murphy (2022), the classification model of LDA is of the form:

$$p(y = c|x, \theta) = \frac{p(x|y = c, \theta)p(y = c|\theta)}{\sum_{c'} p(x|y = c', \theta)p(y = c'|\theta)} \quad (2.19)$$

where the terms,

$p(y = c|\theta)$ is the prior over class labels,

$p(x|y = c, \theta)$ is the class conditional density for class c .

This is a generative model, as it can sample from features x by sampling from the class conditional density. Depending on the choice of class conditionals, the model changes:

- **Gaussian:** generative classifier where the class conditional densities are multi-variate Gaussian distributions
- **Quadratic:** The decision boundary between any two classes, say c and c' , will be a quadratic function of x . Thus the name is defined by the Discriminant Function.

- For a very specific choice of class conditionals, the resulting posterior probability will be a linear function $\log p(y = c|x, \theta = w^T x + \text{const})$ where w is derived from θ . Thus, the overall method is called Linear Discriminant Analysis (LDA).

Linear Decision Boundaries

LDA is a special case of Gaussian discriminant analysis in which the covariance matrices are tied or shared across classes, so $\Sigma_c = \Sigma$. The Discriminant Function is of the form:

$$\log p(y = c|x, \theta) = -\log \pi_c - \frac{1}{2}(x - \mu_c)^T \Sigma^{-1}(x - \mu_c) + \text{const} \quad (2.20)$$

$$= \underbrace{-\log \pi_c - \frac{1}{2}\mu_c^T \Sigma^{-1}\mu_c}_{\gamma_c} + x^T \underbrace{\Sigma^{-1}\mu_c}_{\beta_c} + \underbrace{\text{const} - \frac{1}{2}x^T \Sigma^{-1}x}_{\kappa} \quad (2.21)$$

$$= \gamma_c + x^T \beta_c + \kappa \quad (2.22)$$

Where κ is independent of c , and therefore is an irrelevant additive constant that can be dropped.

Hence we see that the discriminant function is a linear function of x , so the decision boundaries will be linear. Hence this method is called linear discriminant analysis or LDA.

2.4.3 Support Vector Machine (SVM)

The **Support Vector Machine (SVM)** is a supervised learning algorithm widely used for classification tasks Cortes & Vapnik (1995). The key objective of SVM is to find an optimal separating hyperplane that maximizes the margin between data points belonging to different classes. This ensures that the model is not only accurate on the training data but also robust in generalization.

Separating Hyperplane and Maximum Margin

Given a training set $\{(x_i, y_i)\}_{i=1}^N$, with feature vectors $x_i \in \mathbb{R}^d$ and labels $y_i \in \{-1, 1\}$, the SVM seeks a hyperplane of the form $w^T x + b = 0$ that separates the two classes with the largest possible margin.

This margin is inversely proportional to the norm of the weight vector w , and is given by:

$$M = \frac{2}{\|w\|} \quad (2.23)$$

To maximize this margin, the following convex optimization problem is solved:

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to } y_i(w^T x_i + b) \geq 1 \quad \forall i \quad (2.24)$$

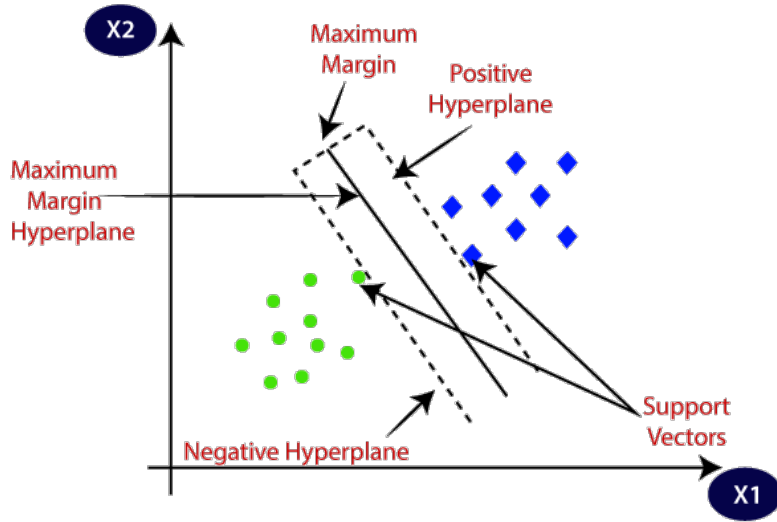


Figure 2.7: Support Vector Machine

where w is a vector normal to the hyperplane, b is the bias term, and the margin is defined as the distance between the closest points of the two classes and the separating hyperplane.

The objective of SVM is to find w and b that maximize the margin while correctly classifying the data.

The optimization problem can be formulated as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad (2.25)$$

subject to the constraints:

$$y_i(w^T x_i + b) \geq 1, \quad \forall i = 1, \dots, N \quad (2.26)$$

This formulation ensures that each data point is correctly classified while maximizing the margin. In real-world scenarios, such as EEG-based stress classification, the data are often not linearly separable in their original feature space. To address this, SVM employs a **kernel trick**, allowing the model to operate in an implicitly transformed higher-dimensional space without explicitly computing the transformation. The transformation function $\phi(x)$ is introduced to map the data into a higher-dimensional space:

$$K(x_i, x_j) = \phi(x_i)^T \phi(x_j) \quad (2.27)$$

The **Gaussian kernel** (also called Radial Basis Function, RBF) is particularly effective for EEG signals due to its ability to model complex, non-linear relationships between high-dimensional features:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \quad (2.28)$$

Here, the parameter σ controls the width of the Gaussian and determines how much influence each training example has. A smaller σ leads to more flexible decision

boundaries, while a larger one results in smoother boundaries. This flexibility makes the RBF kernel a strong candidate for capturing subtle patterns in EEG activity related to cognitive states such as stress.

Support vector Classifier (linear SVC) In this work, the linear SVC is implemented with a maximum-margin classifier with a *linear* decision function

$$f(x) = w^\top x + b, \quad \hat{y} = \text{sign}(f(x)).$$

It seeks a hyperplane that separates the classes while maximizing the margin and controlling violations through a soft-margin penalty.

2.4.4 Gaussian Naive Bayes Classifier (GNB)

A technique that has remained popular over the years, it is specially useful when the dimension p of the feature space is high. The Naive-Bayes model assumes that given a class $G = j$, the features X_k are independent:

$$f_i(X) = \prod_{k=1}^p f_{ik}(X_k) \quad (2.29)$$

While this assumption is generally not true, it does simplify the estimation dramatically:

- The individual class-conditional marginal densities f_{jk} can each be estimated separately using one-dimensional kernel density estimates. This is in fact a generalization of the original naive Bayes procedures, which used univariate Gaussians to represent these marginals.

If a component X_j of X is discrete, then an appropriate histogram estimate can be used. This provides a seamless way of mixing variable types in a feature vector.

Despite these rather optimistic assumptions, naive Bayes classifiers often outperform far more sophisticated alternatives. Although the individual class density estimates may be biased, this bias might not hurt the posterior probabilities as much, especially near the decision regions. In fact, the problem may be able to withstand considerable bias for the savings in variance such a “naive” assumption earns.

2.5 Artificial Neural Networks (ANNs)

An Artificial Neural Network (ANN) is a system for processing information or signals, consisting of numerous interconnected processing elements. These elements work together through direct connections to perform parallel distributed processing, aiming to solve specific computational tasks. Neural networks are inspired by the functioning of biological nervous systems, particularly the human brain, and learn from examples. Compared to conventional computing, ANNs offer a different problem-solving approach. Traditional computer systems rely on algorithms, following a predetermined set of instructions to solve problems. This limits their

ability to solve problems only within the realm of what is already understood and known. In contrast, neural networks and conventional algorithmic computing are not competitors, but rather complement each other. Some tasks, such as arithmetic operations, are better suited for algorithmic approaches, while others benefit from the capabilities of neural networks Macukow (2016). An ANN is constructed using individual units known as artificial neurons or processing elements (PE). These units are interconnected through weighted coefficients, forming the neural structure organized in layers. The strength of neural computations is based on the interconnections between neurons in the network. Each PE receives weighted inputs, applies a transfer function, and produces an output. The behavior of a neural network is influenced by the transfer functions of its neurons, the learning rule employed, and the network architecture itself. The adjustable weights serve as parameters, making a neural network a parameterized system. The activation of a neuron is determined by the weighted sum of its inputs. Multi-Layer Perceptrons (MLPs) are a type of Artificial Neural Network (ANN) characterized by their network structure and interconnections between neurons. In addition to the input layer and output layer, MLPs consist of one or more hidden layers. The defining feature of MLPs is the full connectivity between neurons within and across these layers. An MLP is composed of multiple layers, with each layer consisting of a set of neurons. These neurons are fully connected to both the preceding and succeeding layers. This means that every neuron in a given layer is connected to every neuron in the preceding and subsequent layers, forming a dense interconnection pattern. This full connectivity allows information to flow and propagate through the network, enabling complex computations and learning processes. The input layer serves as the entry point for data, receiving input signals or features. The hidden layers, positioned between the input and output layers, perform intermediate computations and extract increasingly abstract representations of the input data. Finally, the output layer produces the desired output or prediction based on the learned relationships within the network. The connectivity pattern of MLPs plays a crucial role in their ability to model complex relationships and learn from data. By establishing connections between every neuron in adjacent layers, they're able to capture intricate dependencies and non-linear transformations. This flexibility enables MLPs to tackle a wide range of tasks, including classification, regression, and pattern recognition. The depth and width, determined by the number of hidden layers and neurons within each layer, respectively, are design choices that impact the network's capacity to learn and generalize from data. Deeper networks with more hidden layers can capture increasingly abstract and hierarchical representations, while wider networks with more neurons per layer can potentially learn more complex decision boundaries. MLPs, with their multi-layer structure and dense interconnections, have proven to be powerful models for various machine learning tasks. Through training algorithms based on back-propagation, MLPs can adjust the weights and biases associated with each neuron to optimize their performance and learn from labeled examples. The combination of fully connected layers and non-linear activation functions allows MLPs to model highly nonlinear relationships between inputs and outputs, making them versatile and widely used in various domains of machine learning and artificial intelligence. Formally a standard feed-forward Multi Layer Perceptron (MLP) can be defined as follows:

1. Neuron Activation:

$$a_j^{(l)} = f \left(\sum_{i=1}^{n^{(l-1)}} w_{ij}^{(l)} \cdot a_i^{(l-1)} + b_j^{(l)} \right)$$

2. Output Calculation:

$$y_j = f \left(\sum_{i=1}^{n^{(L-1)}} w_{ij}^{(L)} \cdot a_i^{(L-1)} + b_j^{(L)} \right)$$

where:

- $a_j^{(l)}$ represents the activation of neuron j in layer l
- $f(\cdot)$ denotes the activation function applied element-wise to the weighted sum of inputs
- $n^{(l)}$ denotes the number of neurons in layer l
- $w_{ij}^{(l)}$ represents the weight associated with the connection between neuron i in layer $l - 1$ and neuron j in layer l
- $a_i^{(l-1)}$ is the activation of neuron i in layer $l - 1$
- $b_j^{(l)}$ denotes the bias term associated with neuron j in layer l
- y_j represents the output of the MLP for neuron j in the output layer L

It's important to note that the activation function $f(\cdot)$ can vary depending on the specific task and desired properties of the network, such as the sigmoid function ($\sigma(\cdot)$), hyperbolic tangent function ($\tanh(\cdot)$), or rectified linear unit (ReLU) function ($\max(0, \cdot)$). These formulas capture the computation performed by each neuron in the MLP, where the outputs of the preceding layer are combined with weights and biases, passed through the activation function, and propagated forward through the network until reaching the output layer. A graphical representation of a feed-forward MLP can be seen in Fig. 2.8.

Nevertheless, DNNs may have millions of trainable parameters and the small size of the available dataset leads to ineffective learning (Zhu et al., 2019) in Khademi et al. (2023).

2.5.1 Case Study Complexity and Model Selection

The application of Neural Networks to EEG-based Motor Imagery classification is challenged by the limited size and heterogeneity of available datasets. As highlighted in the review by Khademi et al. (2023), shallow Convolutional Neural Networks (brought to fame by Computer Vision) remain viable when turning the input into images, as well as Hybrid models like Zhang's FB-SCP+CNN+LSTM. These two approaches will be studied for completeness.

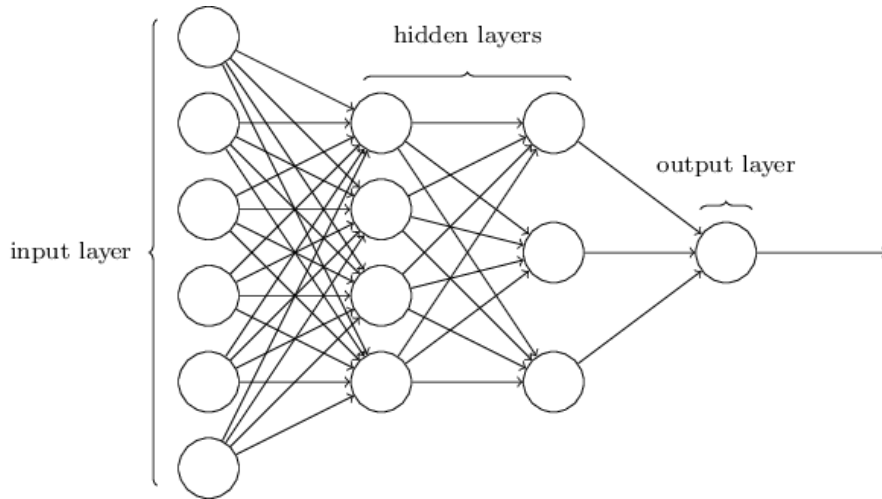


Figure 2.8: Example of a fully connected feed-forward MLP with 2 hidden layers

2.5.2 Convolutional Neural Networks (CNNs)

Convolutional neural networks (CNNs) have garnered remarkable success and emerged as a prominent player in the realm of deep learning. Their contributions to computer vision have paved the way for accomplishing feats previously deemed impossible. Applications such as facial recognition, autonomous vehicles, self-service supermarkets, and intelligent medical treatments have been made feasible through CNN-based computer vision systems Li et al. (2021). The Convolutional Neural Network (CNN) is a type of feedforward neural network that excels in extracting features from data through convolutional structures. Unlike traditional feature extraction methods that require manual intervention, CNN autonomously learns and extracts features. Its architecture draws inspiration from visual perception, where an artificial neuron corresponds to a biological neuron, CNN kernels emulate different receptors capable of responding to diverse features, and activation functions simulate the transmission of neural electric signals surpassing a certain threshold to the next neuron. Additionally, loss functions and optimizers are devised to guide the CNN system in learning our desired outcomes. In comparison to fully connected MLP networks, CNN offers several advantages:

- **Local connections:** Each neuron no longer connects with all neurons in the preceding layer but only to a limited set of neurons. This arrangement effectively reduces the number of parameters, facilitating faster convergence.
- **Weight sharing:** A group of connections can share the same weights, leading to a further reduction in parameters. This sharing mechanism optimizes the network's efficiency.
- **Downsampling dimension reduction:** Through pooling layers, CNN leverages the principle of local correlation in images to downsample and reduce data while retaining essential information. By discarding trivial features, it also helps reduce the number of parameters.

These three characteristics contribute to the widespread recognition of CNN as one of the most prominent algorithms in the field of deep learning. CNN's ability to exploit

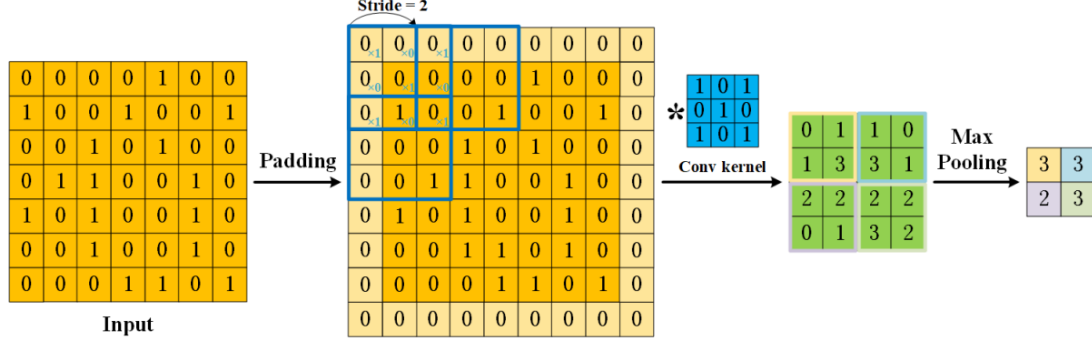


Figure 2.9: Procedure of a 2D convolutional neural network

local connections, weight sharing, and downsampling dimension reduction makes it highly efficient, parameter-efficient, and well-suited for complex data analysis tasks.

Convolution plays a crucial role in the process of extracting features within the context of Convolutional Neural Networks (CNNs). The outputs generated by the convolution operation are commonly referred to as feature maps. However, when applying convolution with a specific kernel size, it can result in the loss of information at the borders of the input data. To address this issue, padding is introduced, which involves enlarging the input by adding zero values (zero-padding method), indirectly adjusting the size of the input. Additionally, the stride parameter is utilized to control the density of the convolution process. A larger stride value leads to a lower density of convolutions. Following the convolution step, the feature maps contain a large number of features, which can potentially contribute to the problem of overfitting. In order to mitigate redundancy and downsize the feature maps, a technique called pooling, also known as downsampling, is introduced. Pooling methods such as max pooling and average pooling are commonly employed. These pooling operations reduce the size of the feature maps while retaining important information. The overall procedure of a CNN is illustrated in Fig.2.9, highlighting the sequential steps of convolution, pooling, and subsequent layers within the network. These operations, combined with other components of the CNN architecture, enable effective feature extraction and learning from input data. Formally the convolution operation involves sliding a kernel over the input feature map to compute the convolved feature map.

$$\text{OutFM}(i, j) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \text{InFM}(i+m, j+n) \times K(m, n)$$

where:

- $\text{OutFM}(i, j)$ represents the value at position (i, j) in the resulting convolved feature map
- $\text{InFM}(i+m, j+n)$ denotes the value at position $(i+m, j+n)$ in the input feature map
- $K(m, n)$ represents the value at position (m, n) in the convolution kernel
- M and N are the dimensions of the convolution kernel

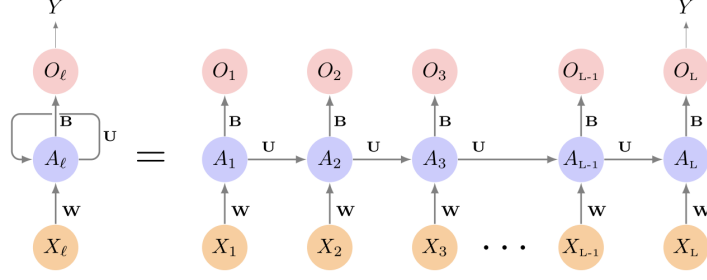


Figure 2.10: Simple RNN

Padding is used to adjust the size of the input feature map before applying the convolution operation. Zero-padding is a commonly used technique.

$$\text{PadFM}(i, j) = \begin{cases} \text{InFM}(i, j) & \text{if } \exists \text{ InFM}(i, j) \\ 0 & \text{otherwise} \end{cases}$$

where:

- $\text{PadFM}(i, j)$ represents the value at position (i, j) in the padded feature map
- $\text{InFM}(i, j)$ denotes the value at position (i, j) in the input feature map

These formulas illustrate the fundamental operations in 2D CNNs, specifically the convolution operation and padding. They form the basis for feature extraction in CNNs and enable the learning of spatially related patterns in 2D data. Generally 2D convolutional neural networks are used in order to extract features which are then flattened and fed to fully connected MLP for the final classification or regression, depending on the task.

2.5.3 Recurrent Neural Networks (RNNs)

EEG data are sequential, in that sense RNNs are uniquely fit for analyzing it James et al. (2023) and Hastie et al. (2017). An RNN is a neural network which maps from an input space of sequences to an output space of sequences. That is, the prediction of output y_t depends not only on the input x_t , but also on the hidden state of the system, h_t , which gets updated over time, as the sequence is processed.

In RNNs, the input object x_t is a sequence. The order of elements and the distance between them carries information in itself. 3.7

3.7 represents a simple RNN, with an input sequence $X = [X_1, X_2, \dots, X_L]$, or also $\{X\}_1^L$; it has a simple output Y , and a hidden layer sequence $\{A_l\}_1^L = [A_1, A_2, \dots, A_L]$. Each X_l is a vector with p elements so $X_l^T = [X_{l1}, X_{l2}, \dots, X_{lp}]$, in the EEG example it would contain several features or signal inputs. Similarly, each A_l is a vector with K elements, so $A_l^T = [A_{l1}, A_{l2}, \dots, A_{lK}]$. At each point, the hidden layer takes input from the current vector X_l and the preceding hidden layer, A_{l-1} ; also at each point, every hidden layer produces an output O_l and feeds into the next output layer A_{l+1} . Then,

$$A_{lk} = g(w_{k0} + \sum_{j=1}^p w_{kj} X_{lj} + \sum_{s=1}^K u_{ks} A_{l-1,s}) \quad (2.30)$$

Where $g(\cdot)$ is a sigmoid activation function like ReLU. And the output O_l in the single output case is given by:

$$O_l = \beta_0 + \sum_{k=1}^K \beta_k A_{lk} \quad (2.31)$$

As we proceed from beginning to end, the activations A_l accumulate a history of what has been seen before, so that the learned context can be used for prediction. The different loss functions depend on the specific problem setting and data.

Chapter 3

Materials and Methods

The Materials and Methods section of this thesis presents a comprehensive description of the experimental setup and procedures employed to investigate the benchmarking of several models. MOABB library was used to create a replicable pipelining for the experiment, this helps with reproducibility as well as streamline the process for future research [Zhu et al. \(2018\)](#). By carefully outlining the materials used and the step-by-step procedures followed, this section aims to ensure transparency, reproducibility, and accuracy in the research process. Furthermore, it provides a foundation for evaluating the reliability and validity of the obtained results.

3.1 Dataset Overview

Dataset for MI benchmarking: BCI Competition IV, dataset 2a from [Tangermann et al. \(2012\)](#). As described by them, All subjects were sitting in a comfortable armchair in front of a computer screen. At the beginning of a trial ($t = 0$ s), a fixation cross appeared on the black screen. In addition, a short acoustic warning tone was presented. After 2 s ($t = 2$ s), a cue in the form of an arrow pointing either to the left, right, down, or up (corresponding to one of the four classes left hand, right hand, foot, or tongue) appeared and stayed on the screen for 1.25 s. This prompted the subjects to perform the desired motor imagery task. No feedback was provided. The subjects were instructed to carry out the motor imagery task until the fixation cross disappeared from the screen at $t = 6$ s. A short break with a black screen followed, a graphical sketch in 3.1.

Below, a summary of the dataset:

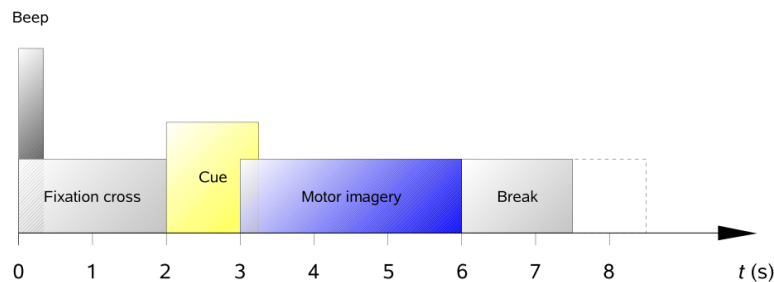


Figure 3.1: Experiment Design from Tangermann et al. (2012)

- Paradigm: 4-class, cue-based motor imagery (MI) task: left hand (1), right hand (2), both feet (3), tongue (4).
- Subjects: 9 (IDs A01–A09).
- Channels: 22 Ag/AgCl electrodes
- Sampling rate: 250 Hz. Band-pass: 0.5–100 Hz. 50 Hz notch* enabled.
- Trials: 62208 trials total:

$$\begin{aligned}
& 144 \text{ trials/class} \times 4 \text{ classes/run} \\
& 576 \text{ trials/run} \times 6 \text{ runs/session} \\
& 3456 \text{ trials/session} \times 2 \text{ session/subject} \\
& 6912 \text{ trials/subject} \times 9 \text{ subjects} \\
& 62208 \text{ trials total}
\end{aligned}$$

A note on **class balance**: As this dataset was created specifically for the BCI Competition IV, it was by design created with class balance in mind, i.e. each class represents 25% of the total; making this a perfectly balanced dataset.

3.2 Experiment 1: Foundation ML Model Pipeline and Baseline

A total of 9 algorithms were benchmarked to choose a baseline of foundational ML models. The basic pipeline was designed as sketched in 3.2. The approach to benchmarking was iterative, the chosen experiment design has two Cross-Validation (CV) steps inside. The "inner" CV is a GridSearch implementation designed to perform hyperparameter tuning, from various candidate hyperparameters taken from the literature. To avoid information leakage, after running it, the only information being used are the hyperparameters. Then, the "outer" CV is a stratified 5-fold, explained at length below, designed to test the proposed hyperparameters. The information for each pipeline can be found in the table below, ordered by the κ results:

Pipeline No.	Preprocessing	Feature Selection	Normalization	Classifier
1	FB + CSP	Y	N	LDA
2	FB + CSP	N	Y	Lin-SVC
3	FB + CSP	Y	Y	Lin-SVC
4	FB + CSP	N	Y	RBF-SVM
5	FB + CSP	Y	Y	RBF-SVM
6	SB + CSP	N	N	LDA
7	SB + CSP	N	N	RBF-SVM
8	SB + CSP	N	N	Lin-SVC
9	FB + CSP	Y	N	GNB

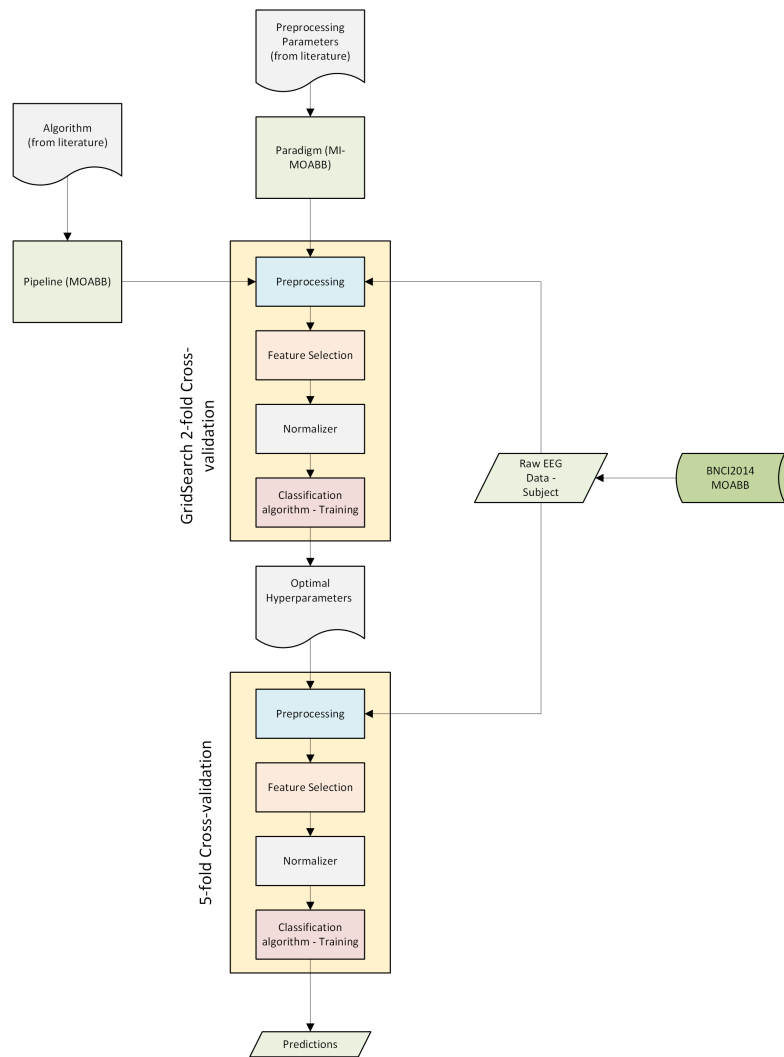


Figure 3.2: Experiment 1 Flowchart

3.2.1 Preprocessing

Filtering

In the context of EEG, filtering is of great importance. It is of interest to the researchers to change the relative amplitudes of the frequency components in signals, or even eliminate them entirely, this is known as filtering Oppenheim et al. (1996).

The first filter applied in the data is a **notch** at 50Hz, a frequency-selective filter that significantly attenuates the electrical artifact present in EEG from the background power-line.

The second is the textbbfband-pass filter: several bands chosen at `fbands = [(8,12),(10,14),(12,16),(14,20),(20,26),(26,30)]`. The values come from the underlying neurophysiological processes described above, specially the observations of the μ and β sensorimotor rhythms and their behaviour in MI.

As per the BCI IV design, the analysis is meant to be causal, since the pipelines are thought to be used online, this rules out zero-phase filtering.

3.2.2 Feature Engineering

Common Spatial Patterns (CSP)

CSP, as mentioned above is a form of spatial filtering, widely used for MI tasks; furthermore it is one of the best performing preprocessing techniques in all the iterations of the BCI competitions, as stated by Tangermann et al. (2012). It is also subject to hyperparameter tuning, with `[4,6,8]` for pipelines with a single band filter and `[2,4,6]` for pipelines with a multi-band pass filter. **Oracle Approximating Shrinkage (OAS)** As there are more channels than the desired spaces or components, CSP performs shrinkage of features, this necessitates the use of particular techniques. OAS is widely used for its computational efficiency and so it has been adopted for this experiment as well. **Log-variance** CSP computes the variance of each CSP component per trial, returns the natural log of that variance (i.e., log-power) as the feature vector. It compresses dynamic range, makes the distribution closer to Gaussian, and balances components—helping linear models like LDA/LinearSVC.

Normalization

The aforementioned Log-variance reporting of CSP is a type of built-in normalisation implemented in the pipeline. However, an extra Normalization step was included into the pipelines for testing.

3.2.3 Model Training and Cross-Validation

Data Splitting

For training, the model is training one subject at a time. The experiment is carried out with MOABB's within-session evaluation, meaning data is split for each session,

which is consistent with the objective of finding generalizable models for online (live) prediction.

Cross-validation is a resampling procedure used to estimate a model’s out-of-sample performance. The data are split into complementary subsets, or folds: the model is trained on $k - 1$ folds and evaluated on the held-out fold; this is repeated so each fold serves once as test, while the other four serve as training data, and the scores are averaged to obtain an unbiased performance estimate. To prevent information leakage, the model is refit (re-trained) every time a new fold is evaluated.

Stratified-CV Is a CV technique that ensures the retaining of the original percentage of labels (in the case of this dataset 25% per class) to avoid overfitting.

In this experiment, the data is split via two Cross-Validation splits.

- **Outer Split:** 5-fold Cross-Validation: implemented to test the best-performing set hyperparameters gotten from the inner CV.
- **Inner Split:** 2-fold Grid-Search Cross-Validation for hyperparameter tuning. Implemented to search through a "grid" of possible combinations for hyperparameters and see which combination works best for each algorithm. As each algorithm will have a different set of hyperparameters to optimize, each pipeline’s grid is described in the table below. The algorithm order is maintained for ease of comparison.

No.	Preprocessing	Feature Selection	Classifier
1	CSP Components [2,4,6]	k-best features [12,24,36]	LDA
2	CSP Components [2,4,6]	k-best features [12,24,36]	SVC c=[0.5,1.0,2.0]
3	CSP Components [2,4,6]	k-best features [12,24,36]	SVC c=[0.5,1.0,2.0]
4	CSP Components [2,4,6]	N/A	RBF-SVM c=[0.5,1.0,2.0] , gamma=["scale",0.05,0.1]
5	CSP Components [2,4,6]	k-best features [12,24,36]	RBF-SVM c=[0.5,1.0,2.0] , gamma=["scale",0.05,0.1]
6	Single Filter Band [(7,13)], [(8,14)], [(9,15)], [(12,18)], [(14,22)], [(16,26)], [(8,30)]] + CSP Components [2,4,6]	N/A	LDA
7	Single Filter Band [(7,13)], [(8,14)], [(9,15)], [(12,18)], [(14,22)], [(16,26)], [(8,30)]] + CSP Components [2,4,6]	N/A	RBF-SVM c=[0.5,2.0,8.0] , gamma=["scale",0.05,0.1]
8	Single Filter Band [(7,13)], [(8,14)], [(9,15)], [(12,18)], [(14,22)], [(16,26)], [(8,30)]] + CSP Components [2,4,6]	N/A	SVC c=[0.25,1.0,4.0]
9	CSP Components [2,4,6]	k-best features [12,24,36]	GNB

Constraints

The dataset’s timing scheme is 7.5s. As previously discussed, the optimal windowing is between 2 and 5 seconds. For this experiment, MOABB’s **default windowing** for the dataset 4s was used with **no overlap**.

Information Leakage

Information leakage can severely affect the generalization ability of models. In order to account for it several features of the design of the experiment were included:

- Nested CV: Inner CV for hyperparameter tuning, no other information leaves the CV. Outer CV for validation of the best performing set of hyperparameters.
- Causal filters to ensure the model can be used in online settings, and at any time, fits on only current and past data.
- No overlap between windows, as in online settings this could not be replicated unless we want to introduce a significant lag in the classification task.

3.2.4 Evaluation

Metrics

For the multi-class MI classification problem attempted in this thesis, Kohen’s Kappa metric will be used, as it best accounts chance correct classification between models; in accordance with the literature.

Kohen’s Kappa (κ): is commonly used for inter-rater reliability for categorical items. Cohen’s kappa measures the agreement between two raters who each classify N items into C mutually exclusive categories:

$$\kappa \equiv \frac{p_0 - p_e}{1 - p_e} = 1 - \frac{1 - p_0}{1 - p_e} \quad (3.1)$$

where p_0 is the relative observed agreement among raters, and p_e is the hypothetical probability of chance agreement, using the observed data to calculate the probabilities of each observer randomly selecting each category. If the raters are in complete agreement (or the classification task is performed perfectly) then $\kappa = 1$. If there is no agreement among the raters other than what would be expected by chance (as given by p_e), then $\kappa = 0$.

3.3 Experiment 2: Artificial Neural Networks

A total of 2 algorithms were benchmarked to compare against the baseline. These are taken from Khademi et al. (2023), and were reproduced from their respective papers:

- Shallow CNN from Xu et al. (2019)

- Hybrid CNN + LSTM(classifier) from Zhang et al. (2021)

3.3.1 Preprocessing

Filtering

Notch: The first filter applied in the data is a **notch** at 50Hz, a frequency-selective filter that significantly attenuates the electrical artifact present in EEG from the background power-line.

Filter Band (FB) The second is FB, several bands chosen at:

`fbands = [(8,12), (10,14), (12,16), (14,20), (20,26), (26,30)]`.

The values come from the underlying neurophysiological processes described above, specially the observations of the μ and β sensorimotor rhythms and their behaviour in MI.

As per the BCI IV design, the analysis is meant to be causal, since the pipelines are thought to be used online, this rules out zero-phase filtering.

3.3.2 Feature Engineering

CNN Feature Engineering

Wavelet transform, as mentioned above is used to turn the EEG signals into images, to pass on to the CNN architecture. With a window for epoching equal to that of the experiment's, i.e. 4s.

RNN Feature Engineering

CSP, as mentioned above is a form of spatial filtering, widely used for MI tasks; furthermore it is one of the best performing preprocessing techniques in all the iterations of the BCI competitions, as stated by Tangermann et al. (2012). It is also subject to hyperparameter tuning, with [4,6,8] for pipelines with a single band filter and [2,4,6] for pipelines with a multi-band pass filter. **Oracle Approximating Shrinkage (OAS)** As there are more channels than the desired spaces or components, CSP performs shrinkage of features, this necessitates the use of particular techniques. OAS is widely used for its computational efficiency and so it has been adopted for this experiment as well. **Log-variance** CSP computes the variance of each CSP component per trial, returns the natural log of that variance (i.e., log-power) as the feature vector. It compresses dynamic range, makes the distribution closer to Gaussian, and balances components—helping linear models like LDA/LinearSVC.

Normalization

`StandardScaler()`: z-normalization, implemented for all ANNs and this helps with training.

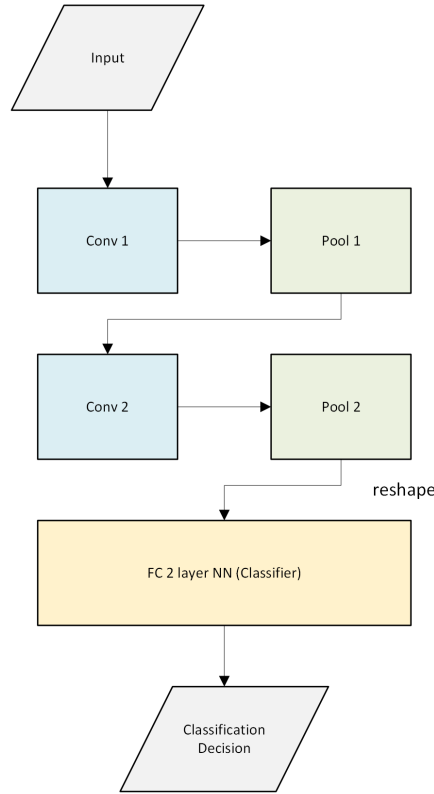


Figure 3.3: End-to-end CNN Architecture

3.3.3 Neural Network Design

CNN end-to-end Design

After the preprocessing and feature engineering, the processed output is passed on to a (Shallow) Convolutional Neural Network, as proposed by Xu et al. (2019), the NN consists of two convolutional steps, reshaping into a 2-layer Fully Connected NN (FC) that culminates in a 4-class classifier, as shown in 3.3.

Hybrid CNN+RNN Design

After the preprocessing and feature engineering, the processed output is passed on to a Hybrid CNN + RNN classifier, as proposed by Zhang et al. (2021), the architecture aims to capture the spatial and temporal information of the EEG signal through a clever design that leverages the strengths of both CNN and RNN architectures. First, the signal is divided into 1s windows, then it is fed to the first CNN, which sports a simple design with only one convolution and no pooling, as shown in ??.

Here, there is a fork, dividing the signal for processing of temporal and spatial features separately. On the spatial side, the output is fed into CNN2, which is the same as the end-to-end but with one extra convolution and pooling step, as shown in 3.5, the output of the CNN2 is the Spatial Feature.

On the side of temporal features, the CNN1 output is reshaped into a sequence that is fed onto the RNN, at each step (1s) the RNN is fed the output of the

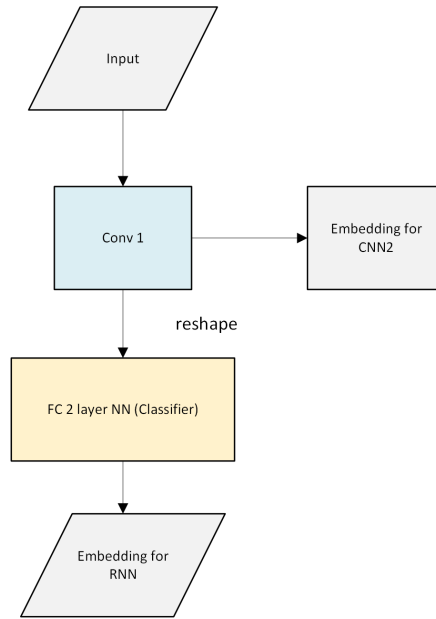


Figure 3.4: CNN1 Architecture

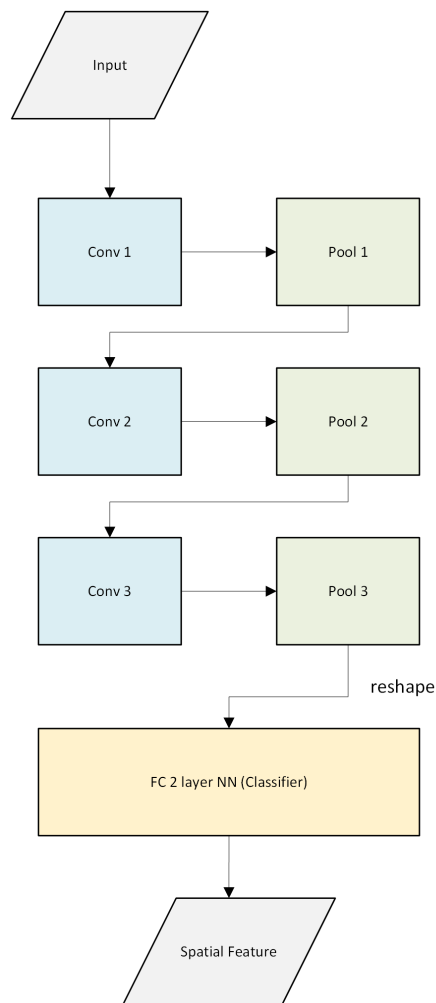


Figure 3.5: CNN2 Architecture

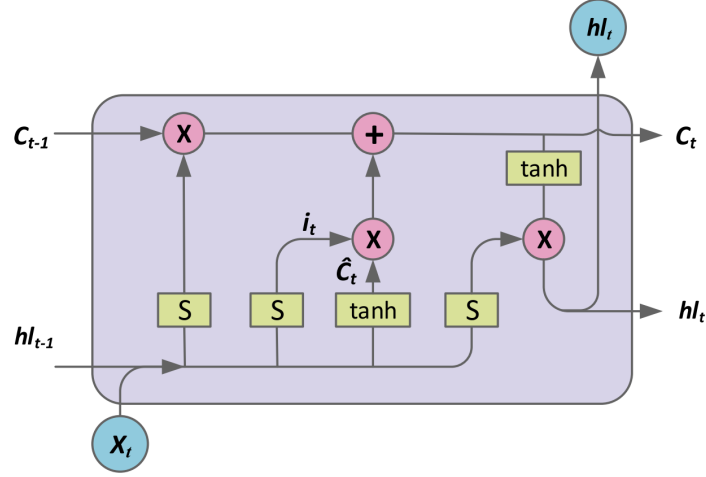


Figure 3.6: LSTM RNN Architecture

corresponding CNN1 time. The RNN is a standard Long Short-Term Memory, as shown in 3.7, the output of this is the Temporal Feature.

Finally, both features get fed onto a Fully connected 2-layer NN that classifies them into one of the four classes, this is the only NN not static during fine-tuning. The complete architecture can be examined in ??.

3.3.4 Model Training and Cross-Validation

Data Splitting

Echoing the first experiment design, the models are training one subject at a time. The experiment is carried out with MOABB's within-session evaluation, meaning data is split for each session, which is consistent with the objective of finding generalizable models for online (live) prediction.

In this experiment, the data is split via two Cross-Validation splits.

- **5-fold Cross-Validation:** implemented to test the best-performing set hyperparameters gotten from the inner CV.
- **Inner Split:** 2-fold hyperparameter tune Cross-Validation for hyperparameter tuning. Implemented to search through a "grid" of possible combinations for hyperparameters and see which combination works best for each algorithm. As each algorithm will have a different set of hyperparameters to optimize, each pipeline's grid is described in the table below. The algorithm order is maintained for ease of comparison

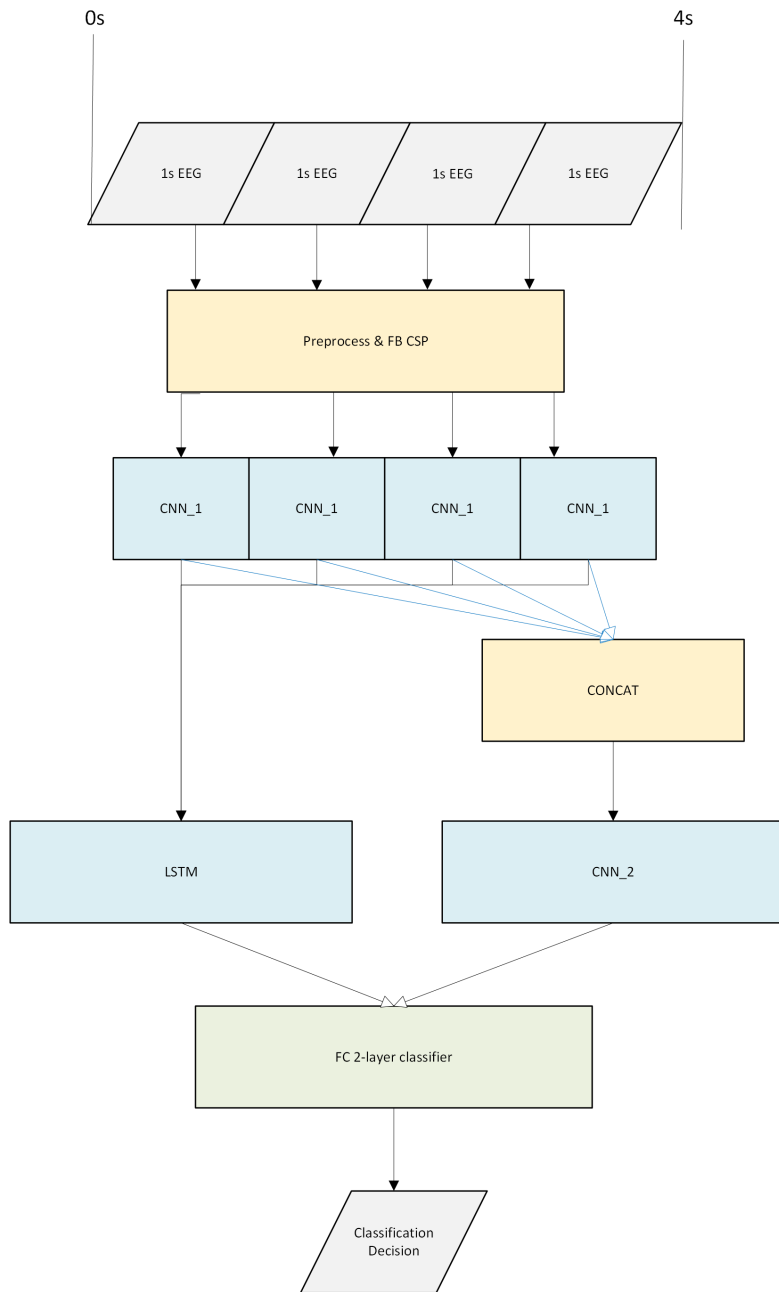


Figure 3.7: End-to-end HDNN Architecture

No.	Preprocessing	NN Features	Classifier
1	Wavelet: morlet vs complex morlet	Kernel 1D vs 2D; NN depth, batch size, epochs	CNN
2	Wavelet: morlet vs complex morlet	Kernel 1D vs 2D, NN depth, epochs, batch size, fine-tuning epochs	Hybrid CNN+RNN+FC

3.3.5 Evaluation

Metrics

Cohen's Kappa κ , will be used to allow for proper comparison between foundational and Deep Learning models.

Chapter 4

Experiment and Results

This section will present the experimental design and compare it among the two experiments.

4.1 First Experiment: Benchmarking Foundational Models

In order to have a baseline for comparison with the cutting-edge ANNs, a first experiment was conducted with the winning algorithms from the BCI IV competition from 2014.

The results are summarized in the table below:

Pipeline	κ	std κ	max_min_diff κ	bal_acc	std_balacc	max_min_diff balacc
Zhang_FBCSP_ML_LDA_CV	0.6481	0.1651	0.4491	0.7361	0.1238	0.3368
Zhang_FBCSP_noFS_LinSVM_CV	0.6361	0.1533	0.3912	0.7270	0.1150	0.2934
Zhang_FBCSP_ML_LinSVM_CV	0.6353	0.1572	0.4028	0.7265	0.1179	0.3021
Zhang_FBCSP_noFS_RBFSVM_CV	0.6289	0.1799	0.4745	0.7216	0.1349	0.3559
Zhang_FBCSP_ML_RBFSVM_CV	0.6253	0.1798	0.4653	0.7189	0.1349	0.3490
Dev_CSP_LDA_CV	0.5741	0.2189	0.6574	0.6806	0.1642	0.4931
Dev_CSP_RBFSVM_CV	0.5720	0.2063	0.6065	0.6790	0.1547	0.4549
Dev_CSP_LinSVM_CV	0.5651	0.2077	0.6389	0.6738	0.1558	0.4792
Zhang_FBCSP_ML_GNB_CV	0.5391	0.1723	0.4815	0.6543	0.1292	0.3611

4.1.1 Confussion Matrices

Here, it is important to also present the confusion matrix for each model, in order to visualize their performance.

From this experiment, 1 baseline was chosen, corresponding to the best performing algorithm of the foundational model round. Zhang's team: Filter Bank + CSP + Select K-Features (MI) + LDA wrapped in the (outer) Cross-Validation, confusion matrix shown in 4.3.

Furthermore, it was proved that the preprocessing step for the pipeline was the same for both models (in line with the literature BCI IV REFERENCE), and that was set as the default for the next experiment (FB + CSP).

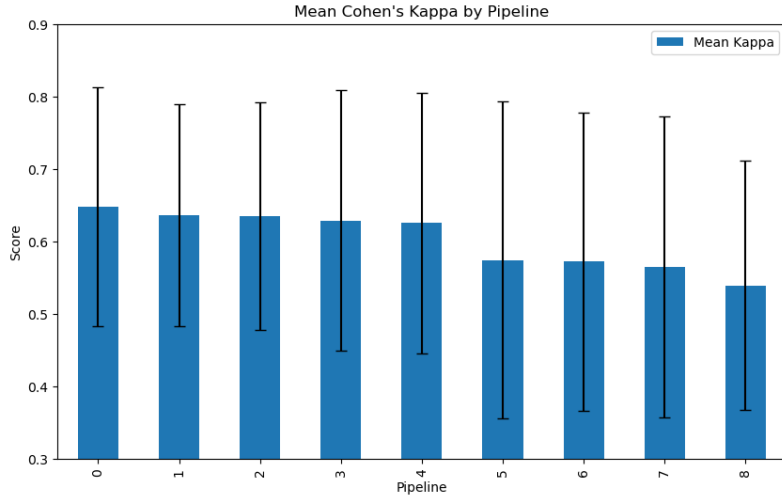


Figure 4.1: Experiment 1 Algorithm Ranking by κ

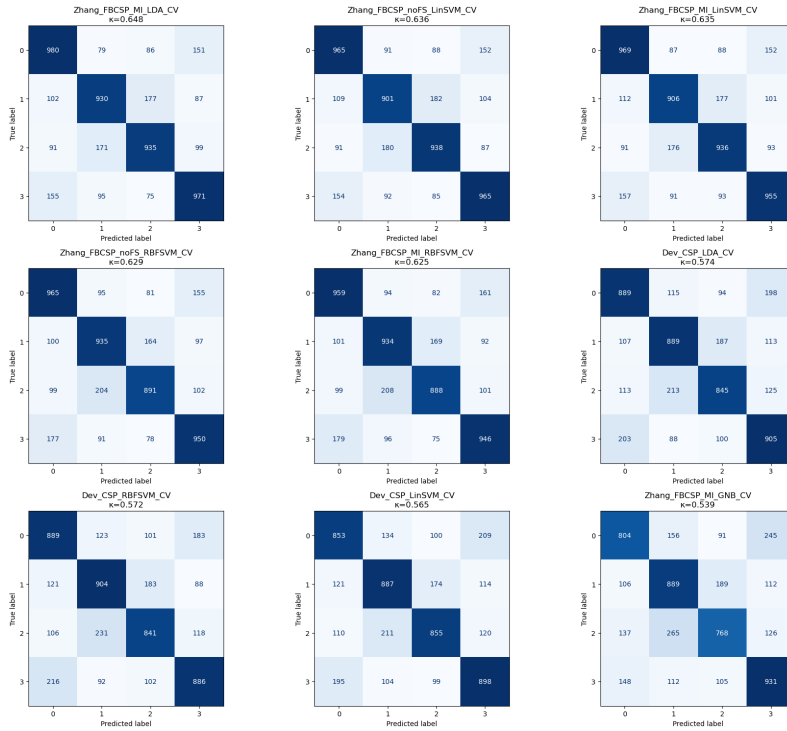


Figure 4.2: Experiment 1 Algorithm Confusion Matrix, ordered by κ

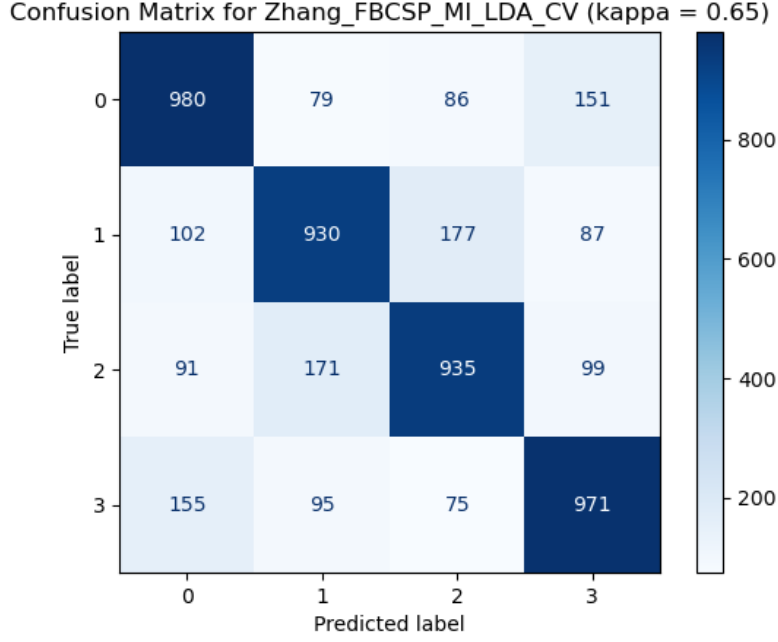


Figure 4.3: Experiment 1 LDA κ

4.2 Second Experiment: ANN Introduction

Once the baseline was set, another experiment was conducted with the ANNs. Two models were tested, replicating the experiments in Xu et al. (2019) and Zhang et al. (2021).

The results are summarized in the table below:

Pipeline	κ	std κ	max_min_diff κ	bal_acc	std_balacc	max_min_diff balacc
Zhang_FBCSP_CNN_LSTM_CV	0.6101	0.16	0.2203	0.7088	0.1502	0.2344
Xu_FT_EEG_MIG_CNN_CV	0.5403	0.1451	0.254	0.6551	0.1432	0.2634

In 4.4 below, we can see the results of the table in graphic form. Lighter shade shows reported κ by classifier from the papers Xu et al. (2019) and Zhang et al. (2021)

Here, it is important to also analyze the confusion matrix for each model, in order to visualize their performance.

4.3 Discussion

While the reported results show that artificial neural networks outperformed the foundational models in terms of accuracy and kappa scores, it is important to note

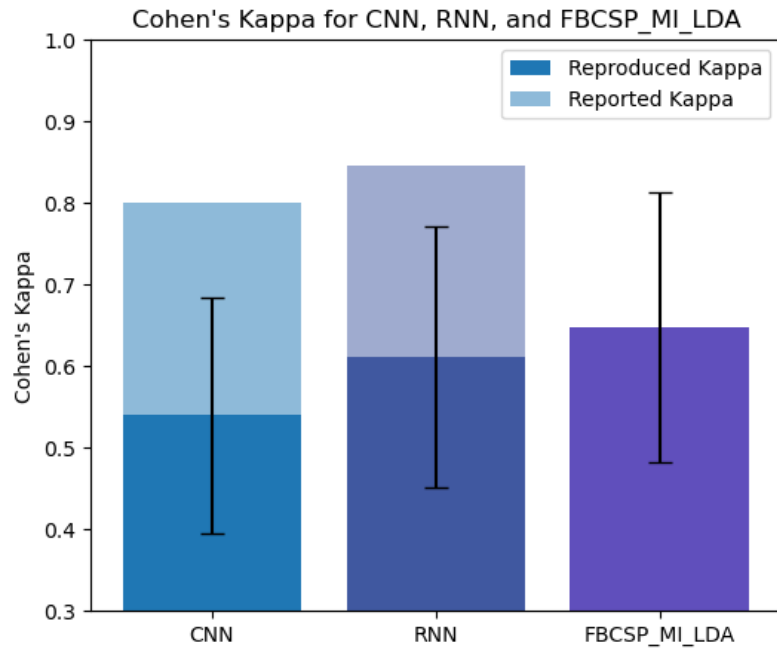


Figure 4.4: Experiment 2 Algorithm ranking, ordered by κ .

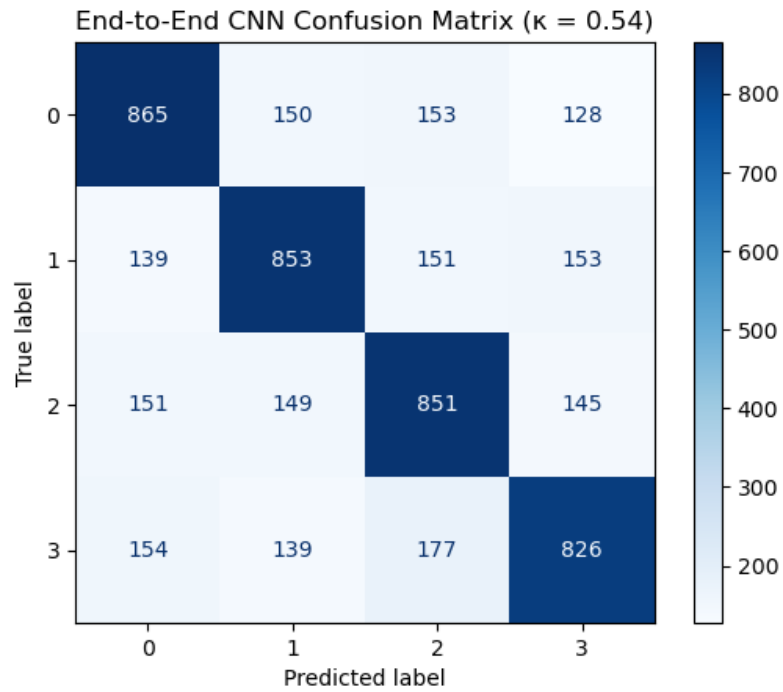


Figure 4.5: Experiment 2: CNN Confusion Matrix

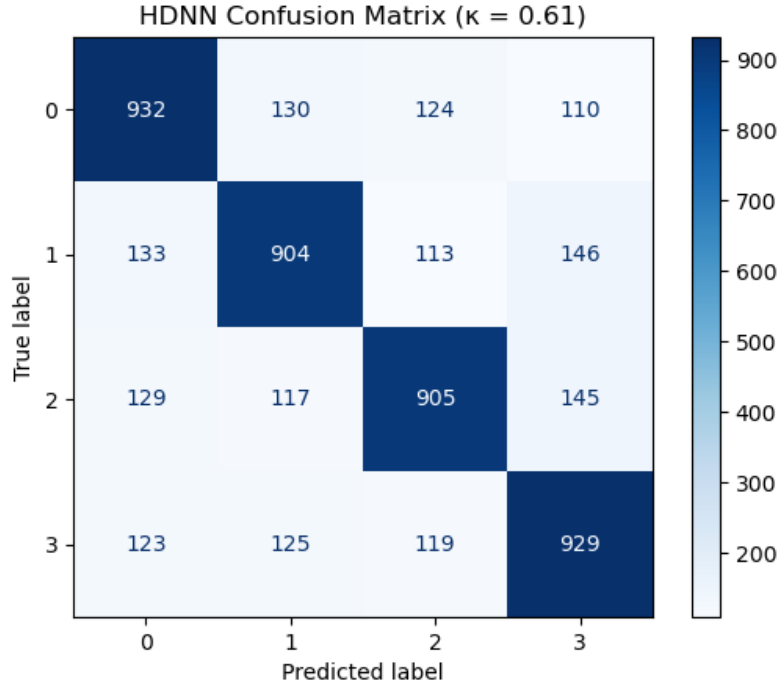


Figure 4.6: Experiment 2: RNN Confusion Matrix

that those results could not be reproduced. In this context, classical approaches remain highly competitive. Methods such as FBCSP combined with LDA or SVM continue to deliver reliable performance with a fraction of the computational cost, training time, and data requirements of deep learning, and were used as preprocessing for the ANNs. This efficiency makes them particularly attractive for real-time and resource-constrained BCI applications, where energy consumption, latency, and portability are critical. Thus, even though ANNs possess a theoretical superior predictive power, foundational models persist as strong baselines that balance interpretability, robustness, and efficiency.

The scope of this thesis was to a) Benchmark foundation models and establish a baseline with which to compare the ANNs, and b) The application of AI techniques to BCI. This undertaking entails the adoption of data-driven statistical models whose accuracy depends on their complexity and the availability of sufficiently large and diverse datasets. The one used for the thesis, albeit large, well designed and openly available, contains only 9 subjects, with very high variability among themselves. This raises the question of wider ability to generalize, and most importantly, the ability of all these models to generalize in "the real world".

The true scope of any BCI experiment is to further the interface of the technologies with people in useful tasks. However, when employed outside of a controlled laboratory setting, BCI applications tend to exhibit severe limitations due to poor out-of-domain data distribution shift robustness. Indeed, the generalization problem appears to be one of the major challenges hindering the deployment of BCIs in the real-world.

In Voluntarily Engaged and Endogenous Tasks, it is essential to ensure that all subjects share the same intent, as variations in cognitive engagement can affect performance (Shifts on subject’s intention). Furthermore, findings from one group of individuals cannot be assumed to generalize to others (Subject shift), nor can they be expected to remain valid when the acquisition device is changed (Acquisition shift) or the BCI system leads to changes in the EEG due to continued use (i.e., neuroplasticity). These challenges underscore a critical limitation, particularly given that most datasets are collected under specific environmental conditions, while the ultimate goal is to develop robust, universal solutions that function reliably in real-world and ecological settings, where various uncontrolled factors may compromise their validity (Shifts on environmental conditions).

Bridging the technological gap: Another key challenge is the computational demand of modern AI models. Most deep learning techniques require substantial processing power, making them impractical for mobile and wearable BCI systems. The integration of Edge AI, which optimizes models for on-device processing, is crucial for enabling real-time, low-latency inference while maintaining power efficiency. Recent efforts to implement AI models on embedded hardware, such as NVIDIA Jetson platforms and low-power AI accelerators, represent promising steps toward overcoming this limitation.

Beyond hardware and computational constraints, there is also a gap in the standardization and reproducibility of EEG analysis. The lack of benchmarking tools and widely accepted datasets makes comparing AI approaches and validating experimental results across studies difficult. Initiatives like the Mother of All BCI Benchmarks (MOABB) Zhu et al. (2018), used in this thesis, are a step toward establishing standardized evaluation protocols, promoting reproducibility, and facilitating the adoption of AI methods in EEG-based BCI research. However, further progress is needed to enhance transparency, encourage data sharing, and improve cross-study comparisons.

4.3.1 Temporal Dynamics of EEGs

One of the fundamental challenges of EEG-based BCI is the accurate modeling of the temporal dynamics inherent in neural activity. EEG signals exhibit highly non-stationary and multi-scale temporal structures, including rapid and transient responses (e.g., ERPs) and sustained oscillatory patterns over different frequency bands Roy et al. (2019). However, most deep learning approaches, including CNNs and RNNs, struggle to fully capture these intricate dynamics because of their inherent limitations in handling long-range temporal dependencies and fine-grained temporal structures.

Chapter 5

Conclusions

The work presented in this thesis set out to benchmark foundational models against artificial neural networks within the context of EEG-based brain–computer interfaces specifically in the realm of MI, with the dual objective of establishing robust baselines and exploring the applicability of modern AI methods to BCI research. By situating these experiments within standardized frameworks such as MOABB and carefully considering both methodological and practical constraints, this thesis contributes to the ongoing dialogue on how best to reconcile accuracy, generalization, and efficiency in neural decoding. The following conclusions synthesize the main findings, reflect on their implications for real-world BCI deployment, and outline potential avenues for future research.

5.1 This research thesis conclusions

It becomes clear that while intricate Deep Learning architectures hold great promise for capturing the inherent temporal and spacial complexity of EEG, foundational models, rooted in classical machine learning, may offer distinct advantages under current constraints. These methods are not only more interpretable and easier to deploy but also significantly cheaper to train and run, requiring orders of magnitude less data and computational resources.

In the context of small, heterogeneous datasets and the practical limitations of mobile BCI systems, as discussed in the previous section, such efficiency is not merely a convenience but a necessity. Thus, foundational models provide a more accessible and sustainable path toward robust BCI solutions, especially when generalization and resource constraints remain critical bottlenecks.

Nonetheless, I believe it is desirable to explore the potential of advanced deep learning techniques for EEG-based BCI. While foundational models remain highly competitive, their simplicity also implies that they may be approaching their theoretical limits in terms of capturing the intricate, nonlinear, and non-stationary dynamics of neural activity. Modern architectures such as recurrent neural networks (RNNs), long short-term memory networks (LSTMs), convolutional neural networks (CNNs), and more recently attention-based transformers offer the capacity to model richer temporal dependencies, hierarchical feature representations, and subject-specific adaptations.

Although these approaches demand larger datasets and greater computational re-

sources, their ability to learn directly from raw or minimally processed signals opens a promising avenue for pushing beyond the ceiling imposed by classical pipelines, potentially enabling more robust and generalizable BCI systems.

5.1.1 Answering the research questions

1. **For Motor Imagery, is there an optimal preprocessing technique for both ML and AI?**

We have explained and explored the three main challenges for MI:

- **Low Signal-to-Noise-Ratio (SNR):** in the literature, it is shown that common spatial pattern (CSP) is an idoneous method to address it, which can extract useful features from the original EEG signal.
- **Non-stationary nature of recorded signals:** addressed by windowing the input fed onto the pipelines as signals over a short time-horizon, exhibit quasi-stationary properties.
- **Variability within subjects:** Addressed by the last pipeline's transfer learning techniques. This approach does not retrain the entire pipeline for each subject, but fine-tunes it as a way to reduce computational costs and increase accuracy.

2. **Which foundational Machine learning models are the best suited ones for classification tasks in Motor Imagery?**

Experiment 1 of the thesis was aimed at answering this question, a selection of the most popular foundational models in MI, was tested. In the end, the **best performing model was the Linear Discriminant Analysis (LDA)**, closely followed by both the linear and Radial Basis Function (RBF) Support Vector Machine (SVM). The difference is sufficiently small that it is not recommended to discard the SVM for classification tasks in MI. As the preprocessing fine-tuning was also not exhaustive, it is possible that a wider grid, other participants or different MI experiment design may favor the SVM (be it linear or otherwise) over LDA.

3. **Are CNNs suited for the BCI-MI Classification problem, and what constraints do they face?** A direct end-to-end CNN implementation, while possible proved to be less than ideal. CNNs excel in image processing, consequently, as shown in the experiment, it is recommended to preprocess the data with a minimal approach and then turn it the signal into an image (following the windowing constraint set by the experiment). Once that is done, CNNs have proven to be a somewhat viable end-to-end solution for MI. It is worth noting however, that in terms of efficiency, they are the worst performing out of the entire set of tested pipelines and so it is not recommended to use them in settings where there are limited computing and power resources (e.g. wearables)
4. **Are RNNs suited for the BCI-MI Classification problem, and what constraints do they face?** A direct end-to-end RNN implementation is possible, and it outperforms the CNN. RNNs, as previously explained excel in

sequence analysis. However, similar to the CNNs, accuracy and overall classification performance can be greatly improved by implementing preprocessing techniques. Moreover, a hybrid approach like the one replicated, originally proposed by Zhang et al. (2021) seems to be pushing the limits of both efficiency and performance, carrying out the task with less resources and greater accuracy.

5.2 Further research

As mentioned above, Transformer-based models, including LLMs adapted to time series data, have demonstrated strong feature extraction capabilities but remain limited by their dependence on large-scale training data and computationally expensive self-attention mechanisms, which may not efficiently capture the sparse, event-driven nature of EEG Ren et al. (2021) signals. Moreover, the current experiments retrained all pipelines for each subject, it may be of interest to prove a different evaluation paradigm in order to streamline processes and make the training phase more efficient. The Hybrid DNN pipeline tested is a promising start in the exploration of ANNs in BCI for the MI context.

Some have proposed biologically inspired neural networks that learn without supervision—such as Adaptive Resonance Theory (ART) ? aiming to achieve autonomous adaptive intelligence. These methods seek to explain how the brain self-organizes representations while adapting to new information in a dynamic environment, a fundamental capability for addressing the generalization challenges of current BCI approaches. In a different vein, spiking neural networks (SNNs) represent a significant alternative in the artificial intelligence landscape, particularly for their application to BCI systems using EEG signals Wang et al. (2020). Unlike traditional artificial neural networks, which rely on continuous activation functions, SNNs process information using discrete spikes that more closely emulate biological neurons’ behavior. This spike-based paradigm increases the biological plausibility of artificial networks. It enables more efficient integration with neuromorphic hardware platforms with significantly lower power consumption than traditional hardware platforms such as GPUs and CPUs Kumar, Tang & Yoo (Kumar et al.). Low power consumption is essential for portable devices, such as BCI wearables or mobile diagnostic tools, which often run on battery power. The increased energy range allows these systems to operate for extended periods without recharging, making them ideal for mobile real-time applications.

Expanding the benchmarking to Transformers based architecture, and SNNs, as well as exploring ways to optimize the classification task in order to avail the use of the pipelines on wearables or resource-limited mobile devices would be very attractive avenues for new research.

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